

Basic Principles of Medicine II

Module: Nervous System and Behavioral Sciences | Lecture NB 18

Oral Cavity & Palate

Dr. Stephen Onigbinde

sonigbin@sgu.edu

Department of Anatomical Sciences

School of Medicine, St. George's University



Copyright

All year-1 course materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed, or published in any form, printed, electronic, or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

'View only' files, such as lecture recordings, are explicitly excluded from download, and creating copies of these recordings by students and other users is strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St. George's University. Users of this document identifying potential violations of these regulations are asked to bring their concerns to the attention of the author.

Recommended reading

Drake RL, Vogl AW, Mitchell AWM. Gray's anatomy for students. 5th ed. Elsevier. Chapter 8.



Pay special attention to the 'In the Clinic' boxes which contain the clinical correlates for this topic.



Objectives

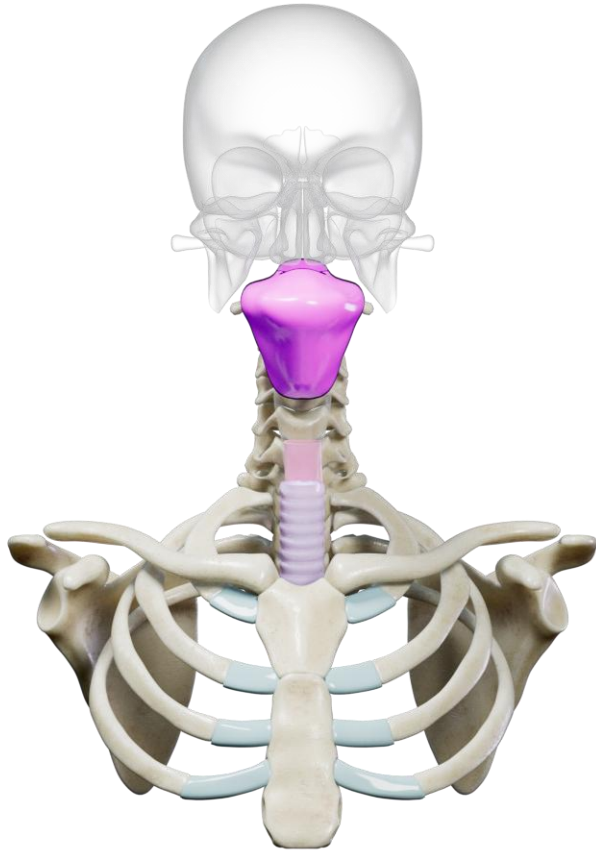
SOM.MKII.BPM2.1.NB.1.ANAT.0109	Describe the boundaries of the oral cavity and its division into the oral cavity proper and oral vestibule
SOM.MKII.BPM2.1.NB.1.ANAT.0110	Describe the sensory innervation of the oral vestibule
SOM.MKII.BPM2.1.NB.1.ANAT.0111	Describe the sensory innervation of the oral cavity proper
SOM.MKII.BPM2.1.NB.1.ANAT.0112	Describe the course of the submandibular duct and its termination under the tongue
SOM.MKII.BPM2.1.NB.1.ANAT.0113	Describe the location of the sublingual gland and the openings of this gland
SOM.MKII.BPM2.1.NB.1.ANAT.0114	Differentiate between the oral and pharyngeal parts of the tongue
SOM.MKII.BPM2.1.NB.1.ANAT.0115	List the nerves that carry general sensation and taste sensation from the tongue
SOM.MKII.BPM2.1.NB.1.ANAT.0116	List the general attachments, innervations and actions of the extrinsic muscles of the tongue
SOM.MKII.BPM2.1.NB.1.ANAT.0117	Describe how the tongue and palate function during deglutition
SOM.MKII.BPM2.1.NB.1.ANAT.0118	List the general attachments, innervations and actions of the muscles that are found in the floor of the mouth
SOM.MKII.BPM2.1.NB.1.ANAT.0119	Describe the arterial supply and venous and lymphatic drainage of the tongue



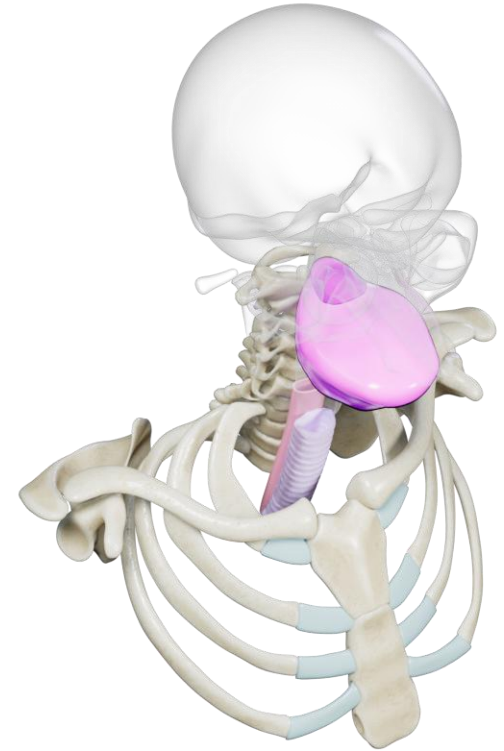
Objectives

SOM.MKII.BPM2.1.NB.1.ANAT.0120	Describe the structure and function of the hard and soft palate
SOM.MKII.BPM2.1.NB.1.ANAT.0121	Describe the bones that form the hard palate
SOM.MKII.BPM2.1.NB.1.ANAT.0122	Describe the mucosal coverings and innervation of the hard palate
SOM.MKII.BPM2.1.NB.1.ANAT.0123	Describe the muscles, function and innervations of the soft palate
SOM.MKII.BPM2.1.NB.1.ANAT.0197	Give the exit foramen for the hypoglossal nerve and state which bone it is found in
SOM.MKII.BPM2.1.NB.1.ANAT.0198	Describe the types of nerve fibers found in the hypoglossal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0199	Describe the branches and distribution of the hypoglossal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0200	Describe the extracranial course of the hypoglossal nerve and what it supplies
SOM.MKII.BPM2.1.NB.1.ANAT.0201	Give the location of the nuclei of the hypoglossal nerve
SOM.MKII.BPM2.1.NB.1.ANAT.0202	Explain the consequence of extracranial lesions of the hypoglossal nerve

Overview



- Oral cavity
- Glands
- Tongue
- Deglutition
- *Cranial nerves: V, VII, IX, X, XII*



 Oral Cavity/ Oropharynx

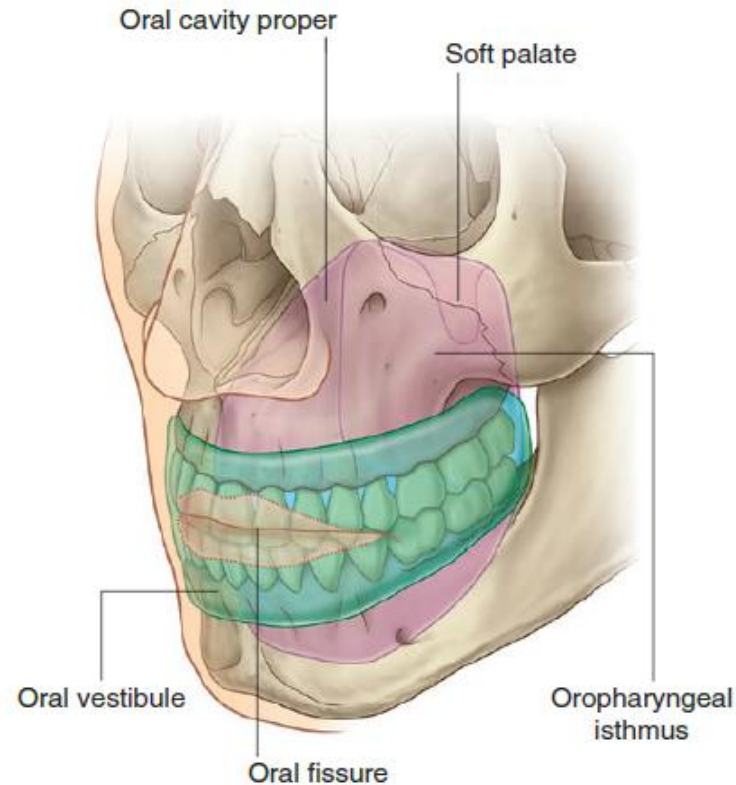
Oral Cavity

- **Oral vestibule**

- Between lips/cheeks and dental arches

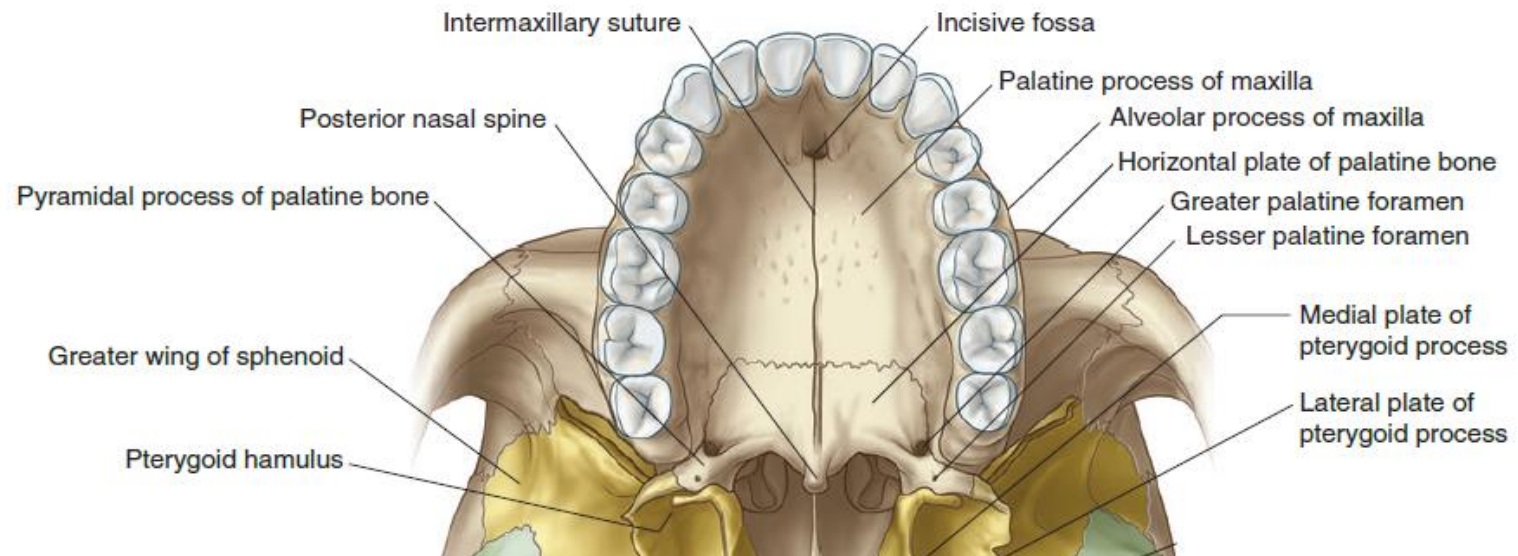
- **Oral cavity proper**

- Between dental arches and *palatoglossal arch*



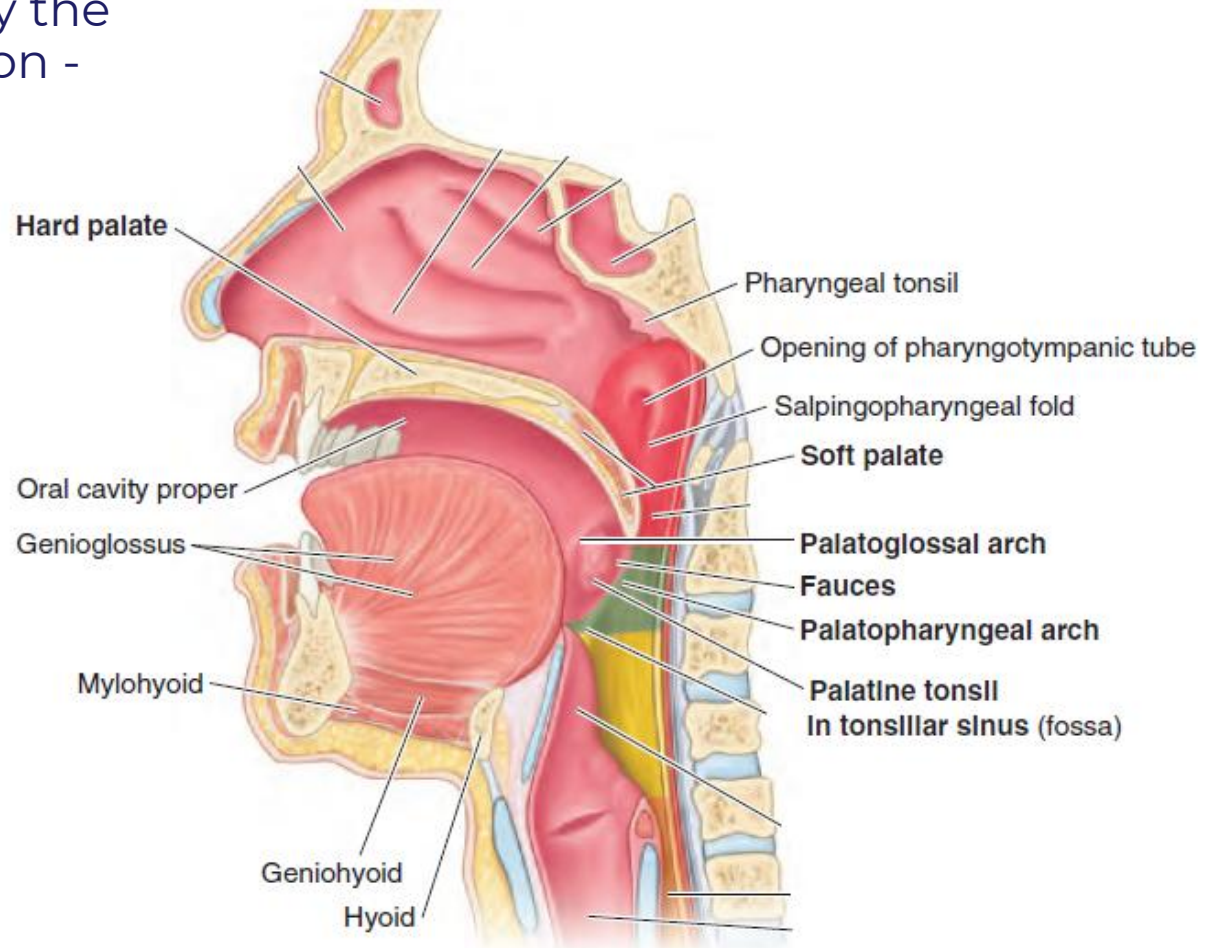
Hard Palate & Bones of the Oral Cavity

- Palate: roof of the mouth
 - separates the oral cavity from the nasal cavity
- Results from fusion at *median palatine suture* of palatine processes of **maxillae** and horizontal plates of **palatine bones**
- Mucosa is tightly bound to underlying bone, with deeper mucus-secreting palatine glands



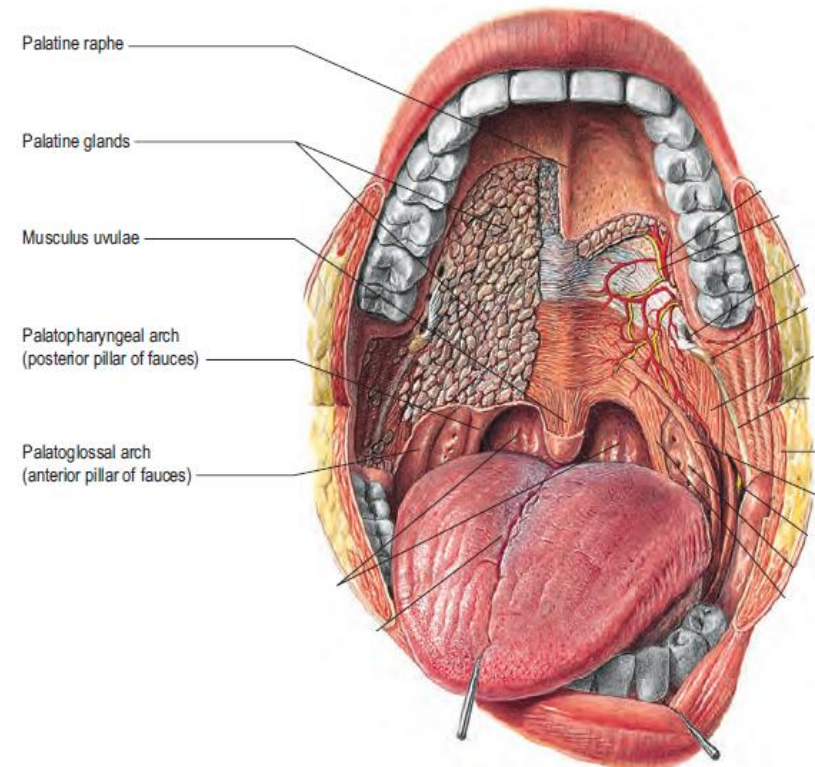
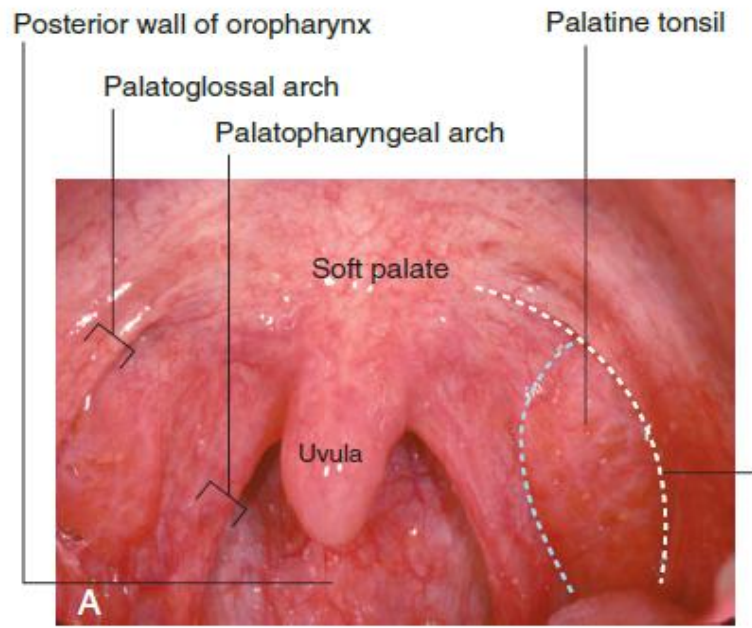
Soft Palate

- Movable posterior third of palate
 - suspended from posterior border of the hard palate by the anterior portion - the *palatine aponeurosis*



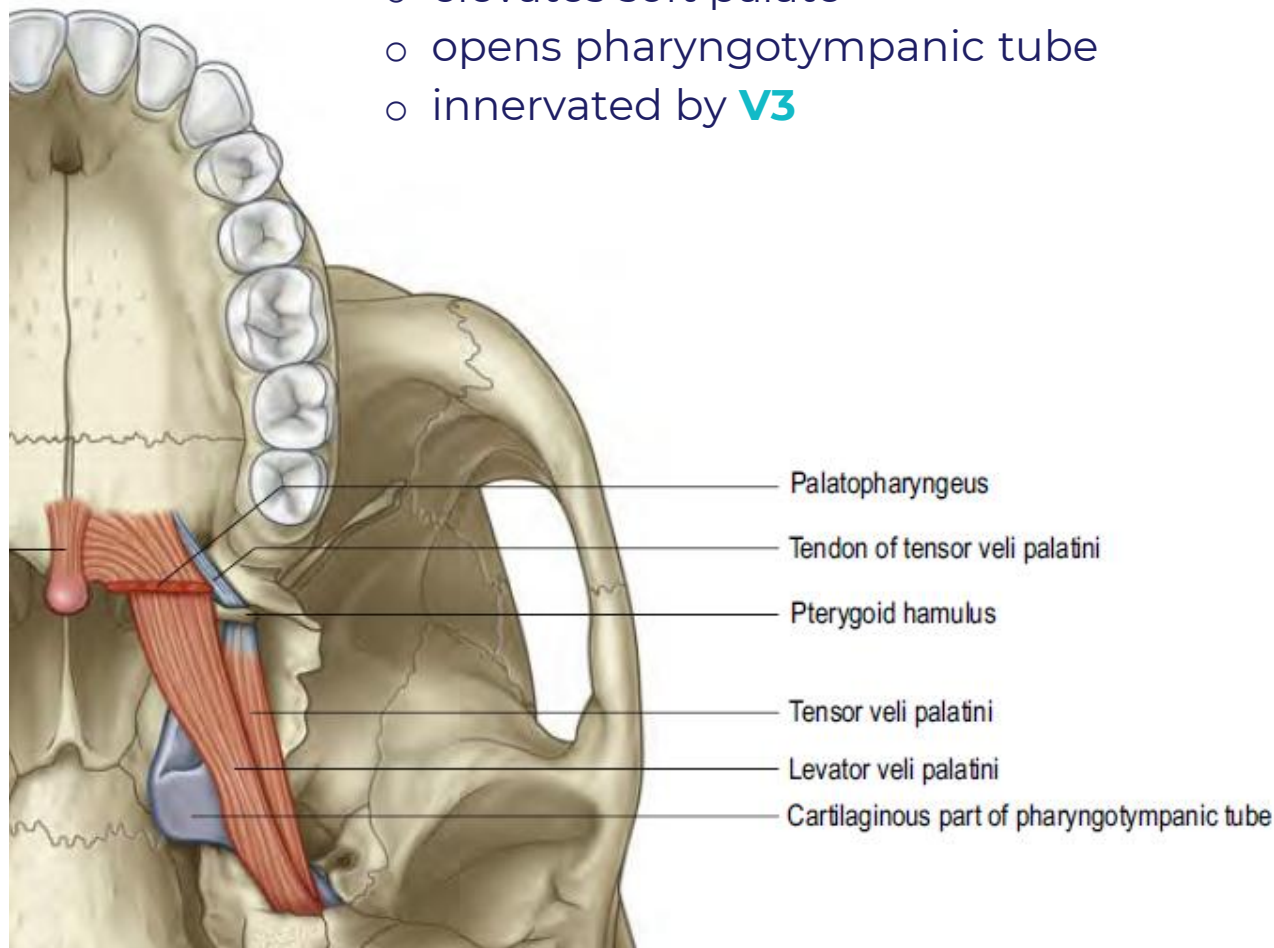
Soft Palate

- **Movable** posterior third of palate
 - muscular part extends posteriorly, from which hangs the **uvula**
 - contains muscle fibers from **musculus uvulae**
 - innervated by **X**
- Fauces: between *palatoglossal arch* and *palatopharyngeal arch*



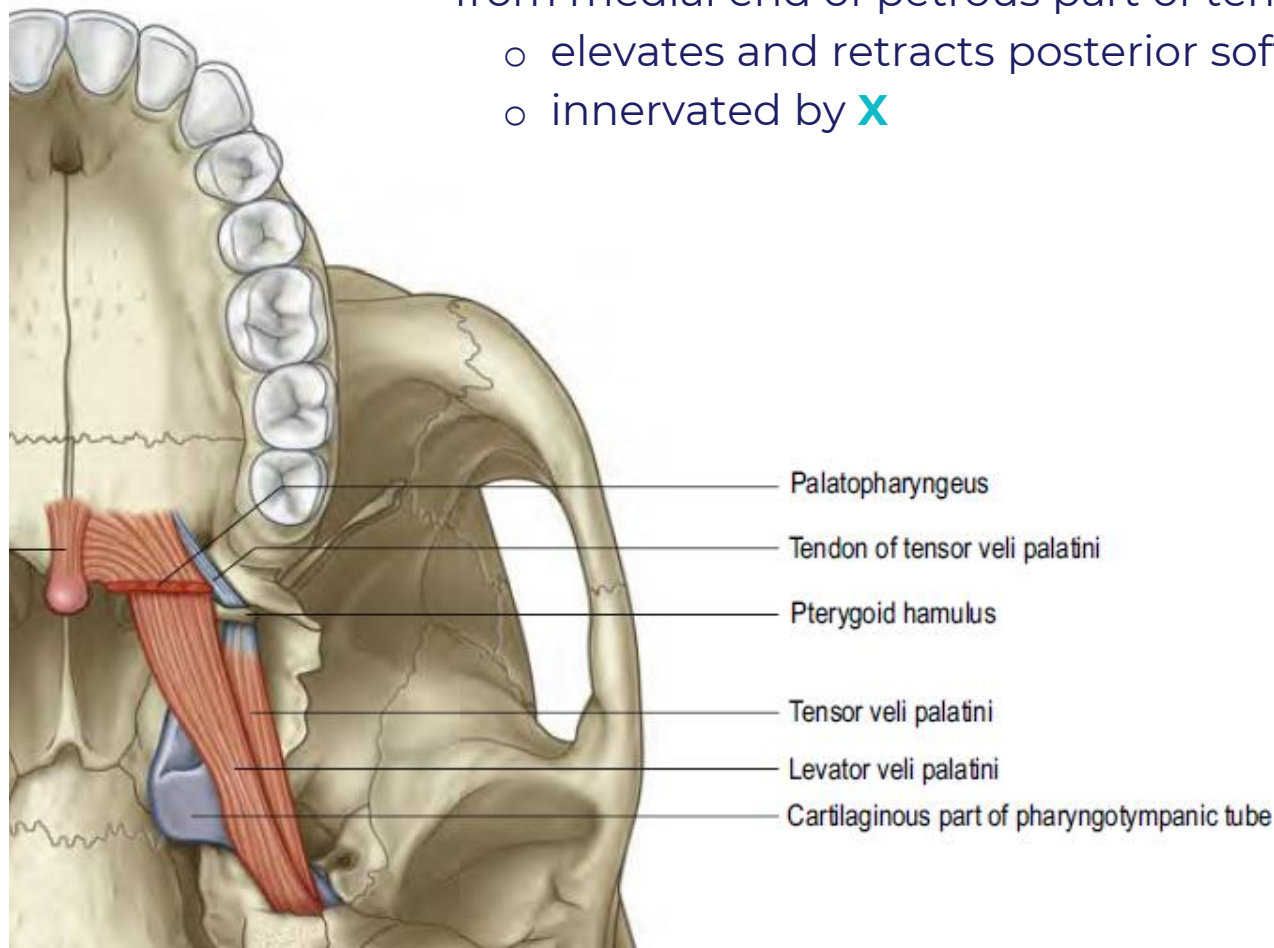
Tensor Veli Palatini

- from fibrous pharyngotympanic tube to palatine aponeurosis
 - elevates soft palate
 - opens pharyngotympanic tube
 - innervated by **V3**



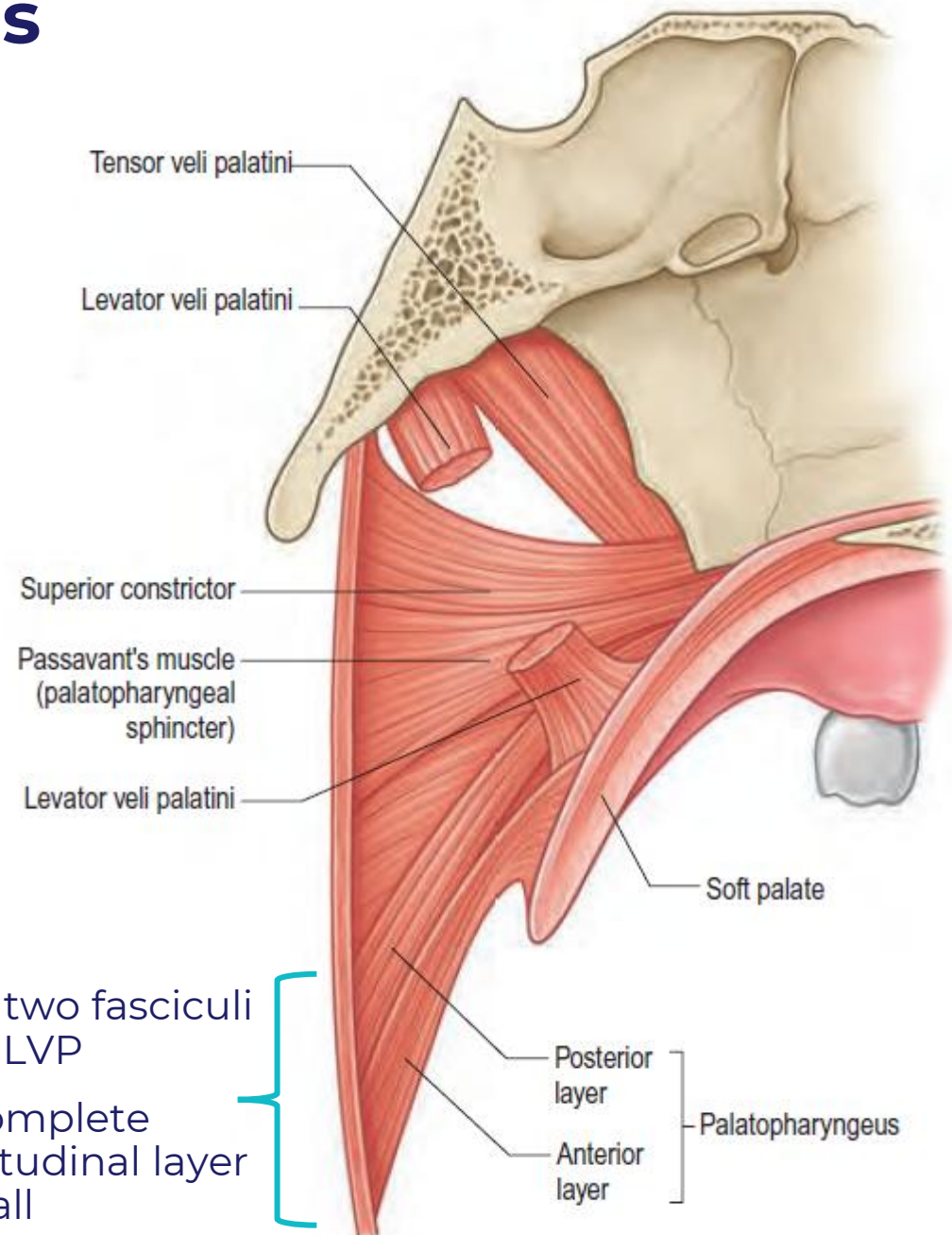
Levator Veli Palatini

- from medial end of petrous part of temporal bone
 - elevates and retracts posterior soft palate
 - innervated by **X**



Palatopharyngeus

- from inferior palatine aponeurosis to lateral tongue
- depresses soft palate
- elevates pharynx
 - innervated by **X**

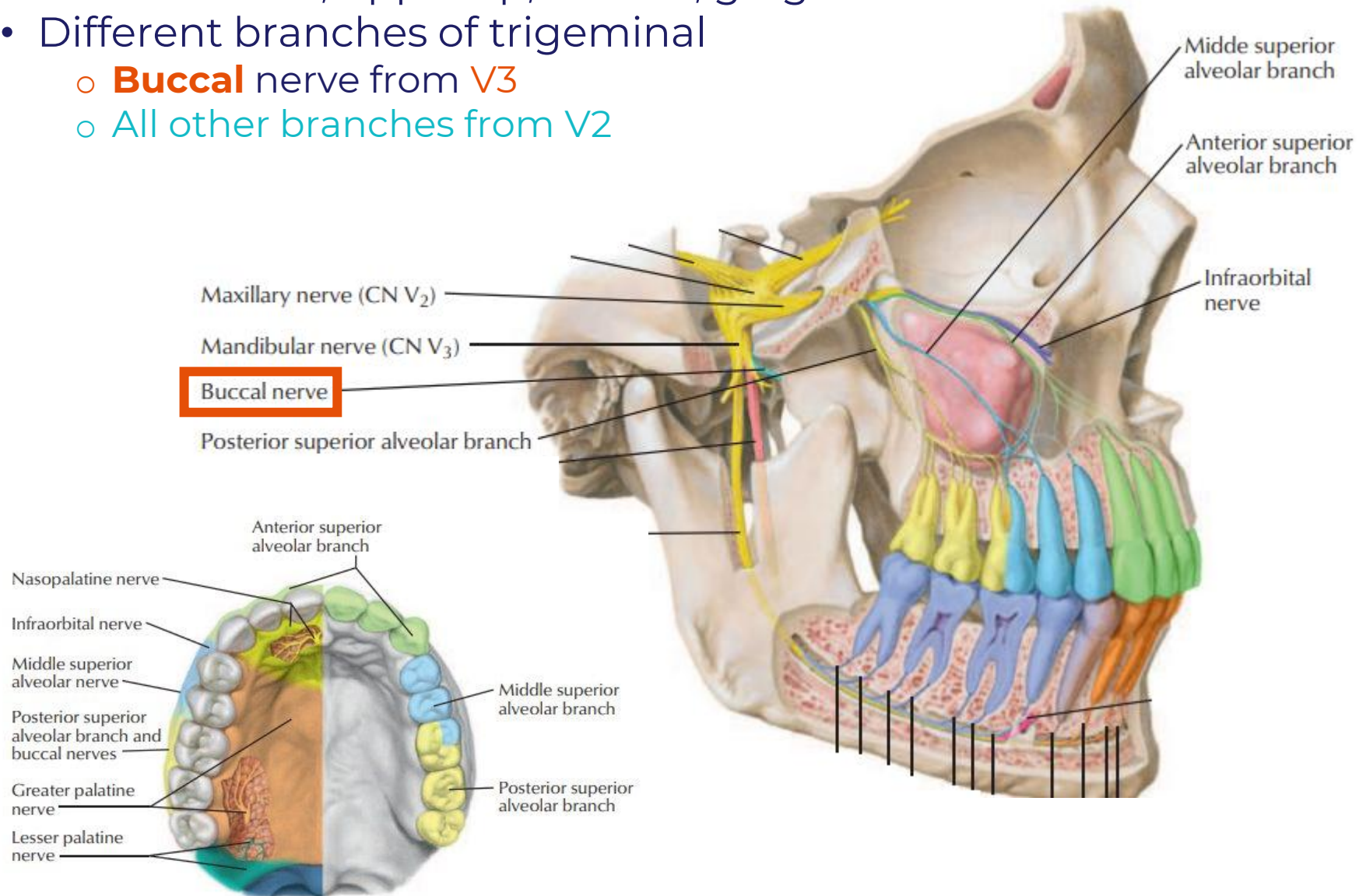


- composed of two fasciculi separated by LVP
- forms an incomplete internal longitudinal layer in pharynx wall

Sensory Innervation to Palate

Palatal mucosa, upper lip, cheeks, gingiva:

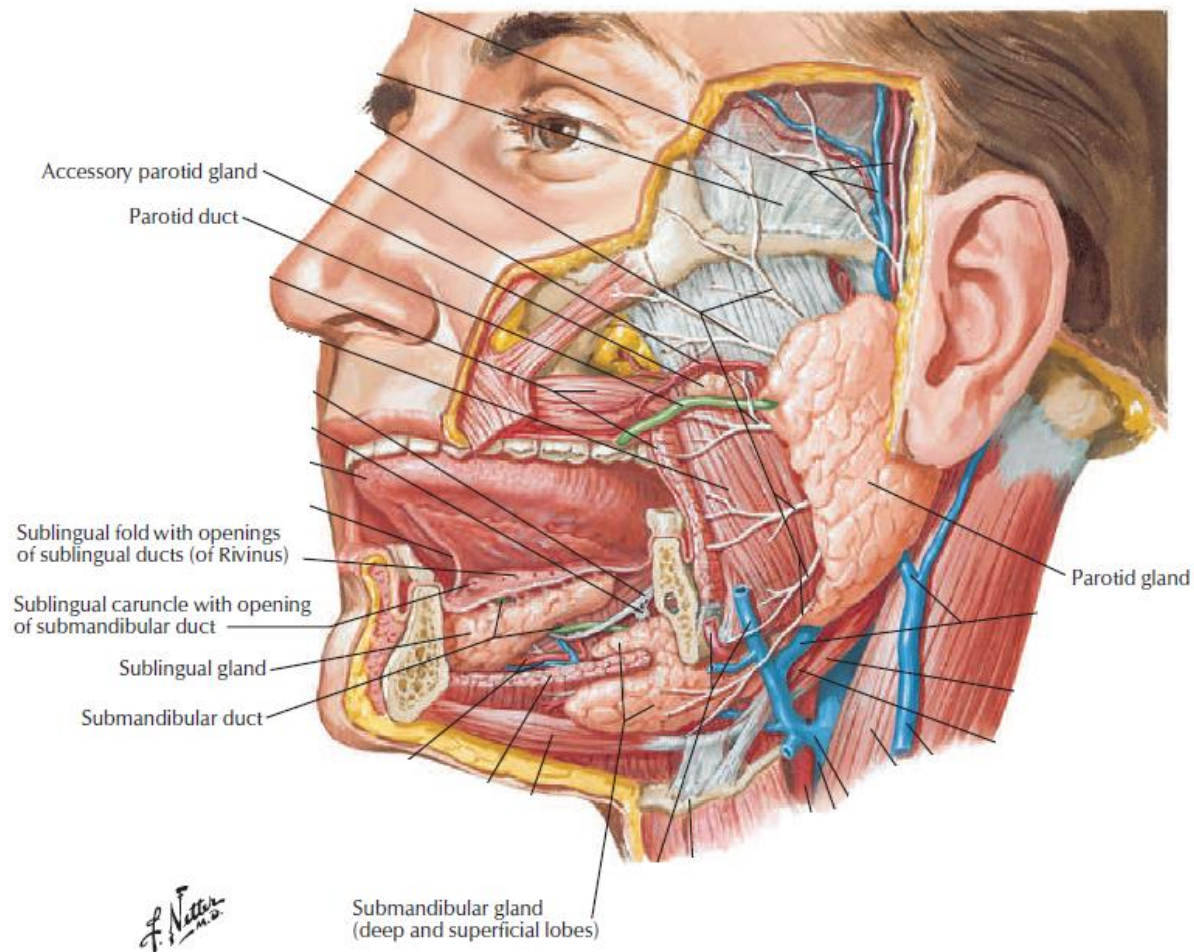
- Different branches of trigeminal
 - **Buccal** nerve from V3
 - All other branches from V2



Salivary Glands

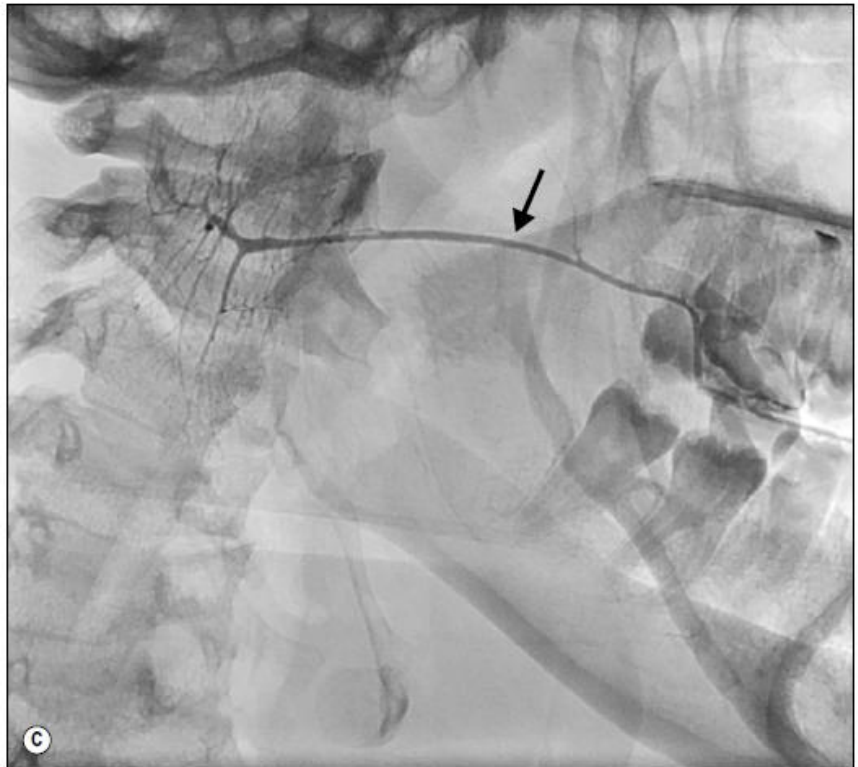
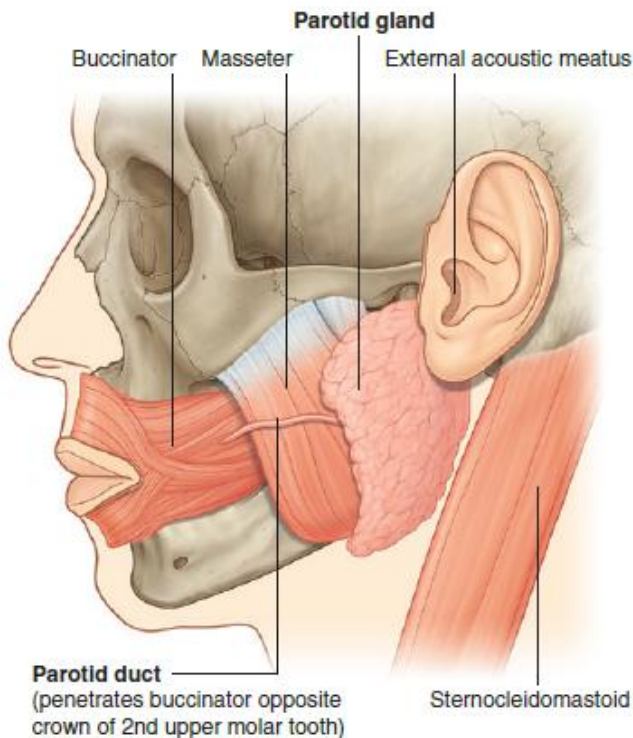
Three large paired glands

- Parotid
- Submandibular
- Sublingual



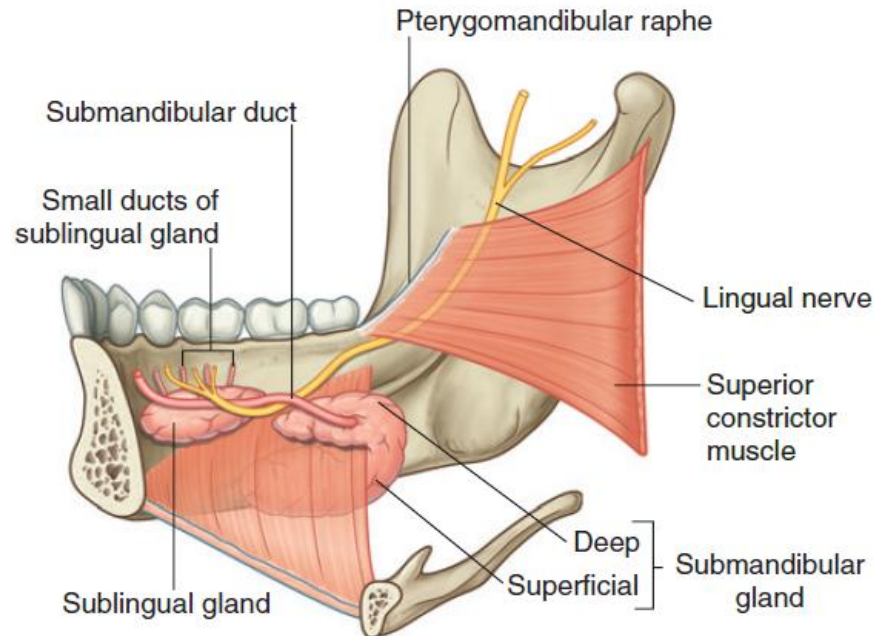
Parotid Gland

- Location:
 - begins posterior to ramus of mandible
 - extend to zygomatic arch
- Parotid (Stensen's) ducts:
 - pierces buccinator
 - opens into oral cavity near second upper molar

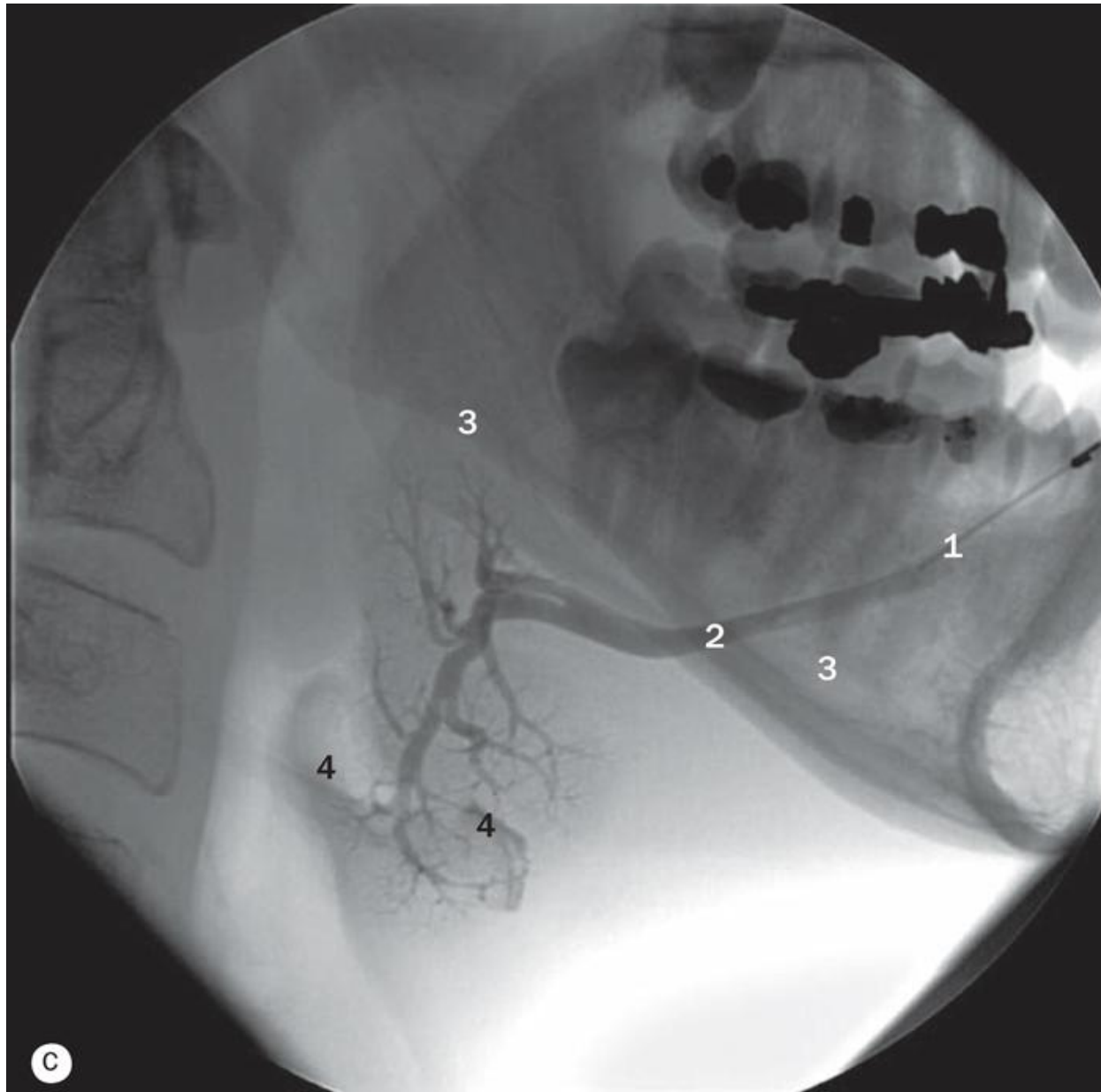


Submandibular Gland

- Location:
 - deep part inside oral cavity
 - superficial part beneath mylohyoid
- Submandibular (Wharton's) duct:
 - emerges from the deep part and runs forward
 - opens at the base of the frenulum
 - double-crossed by the **lingual n**



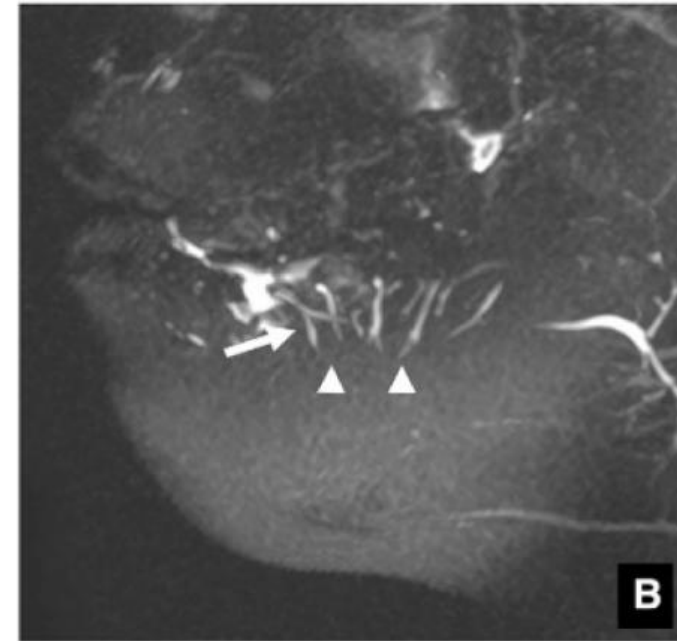
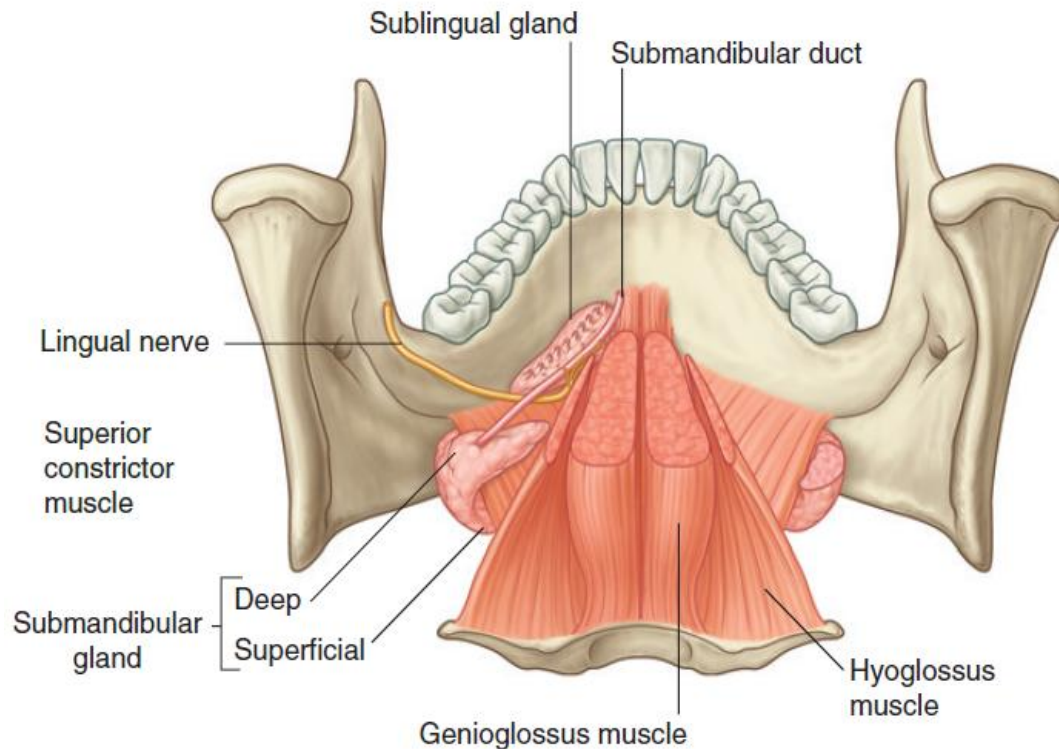
Submandibular Gland Sialogram



Key
1: catheter
2: main (Wharton's) duct
3: mandible
4: secondary ductules

Sublingual Gland

- Location:
 - beneath the mucous membrane
 - along the inner surface of the mandible
- Sublingual ducts (of Rivinus):
 - numerous
 - open onto crest of sublingual fold, into oral cavity

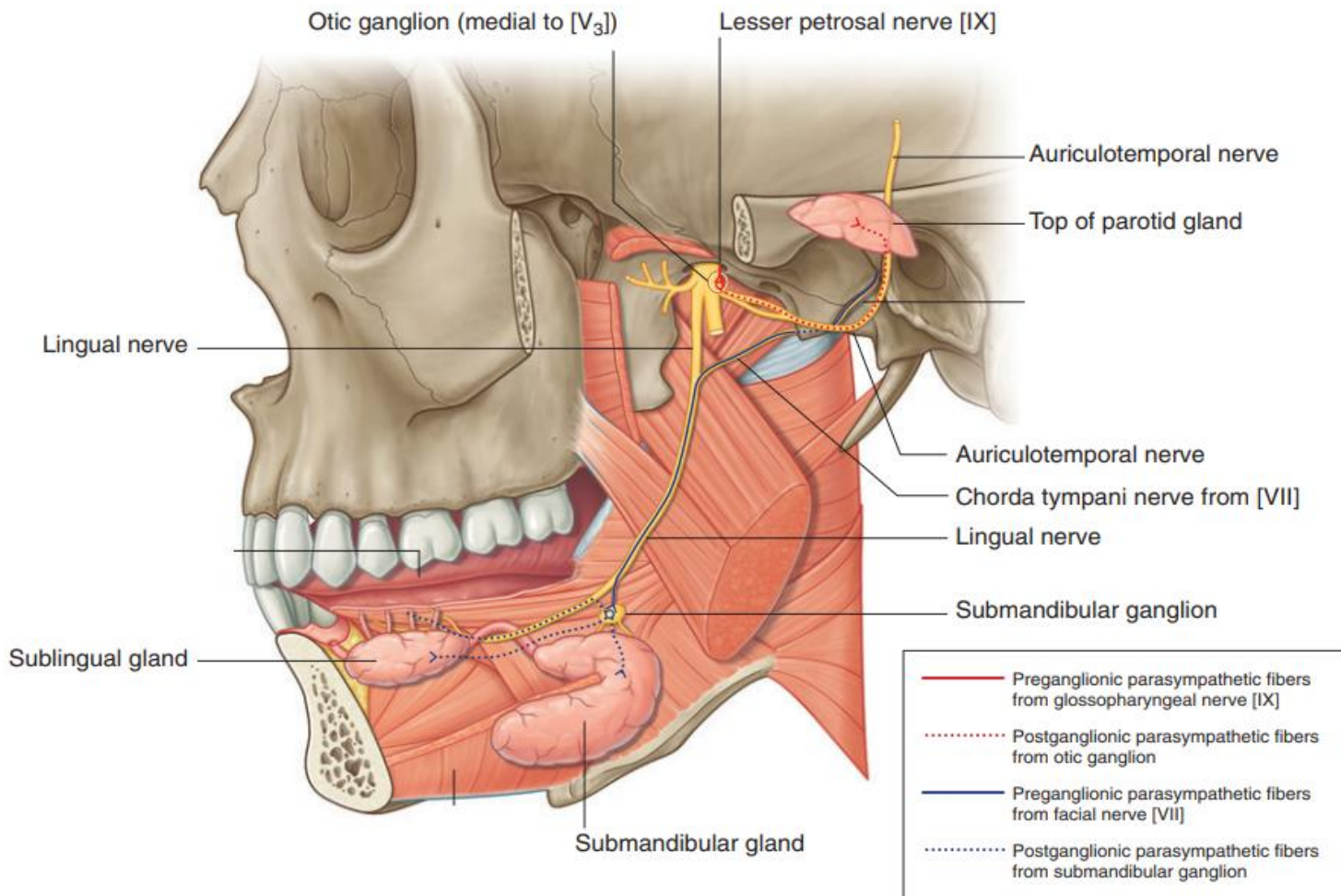


DOI: [10.3390/jcm9113676](https://doi.org/10.3390/jcm9113676)

Innervation of glands – general guide

General guide:

- Above oral fissure (lacrimal, parotid): petrosal nerves
 - **Lacrimal:** greater petrosal
 - **Parotid:** lesser petrosal
- Below oral fissure (submandibular, sublingual): chorda tympani nerve



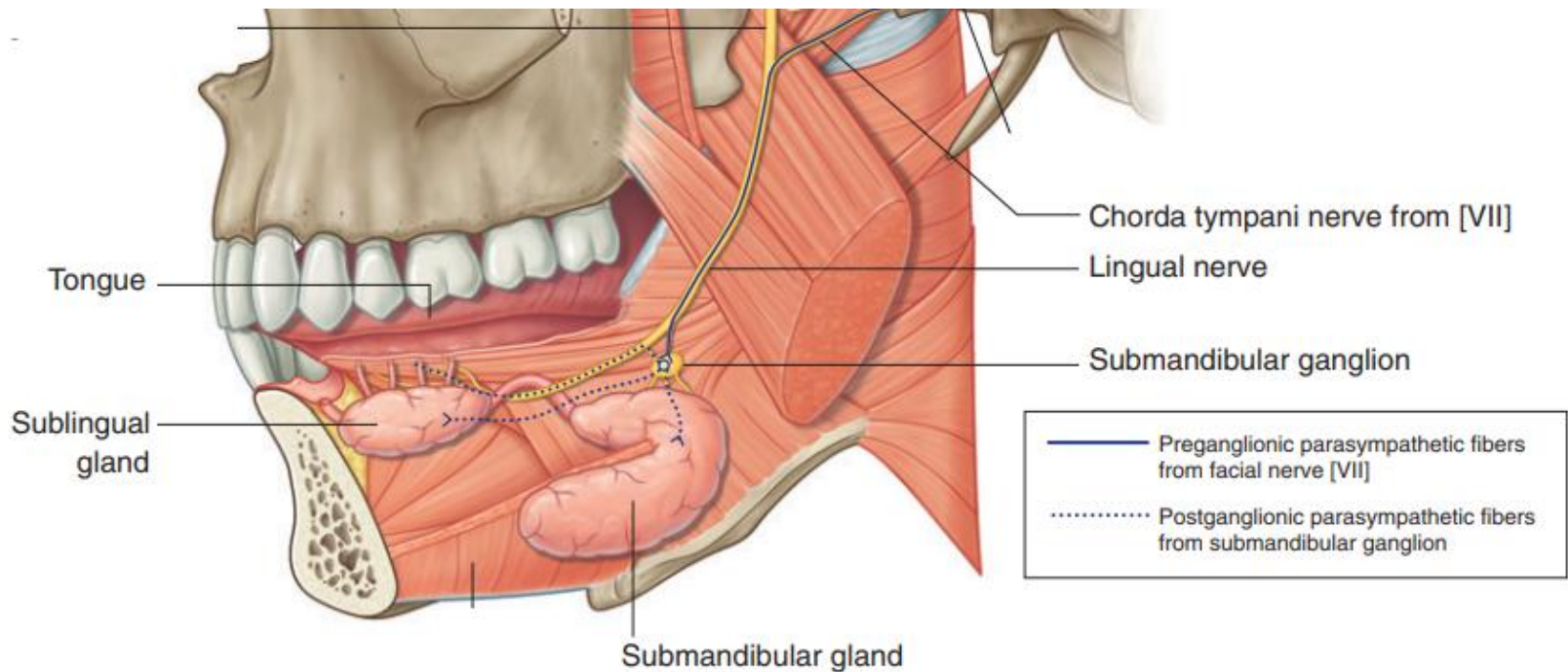
Salivary gland autonomics

Vasomotor

- Sympathetics from superior cervical ganglion

Secretomotor

- Submandibular ganglion
 - posterior filament: preganglionic fibers
 - anterior filament: postganglionic parasympathetic fibers

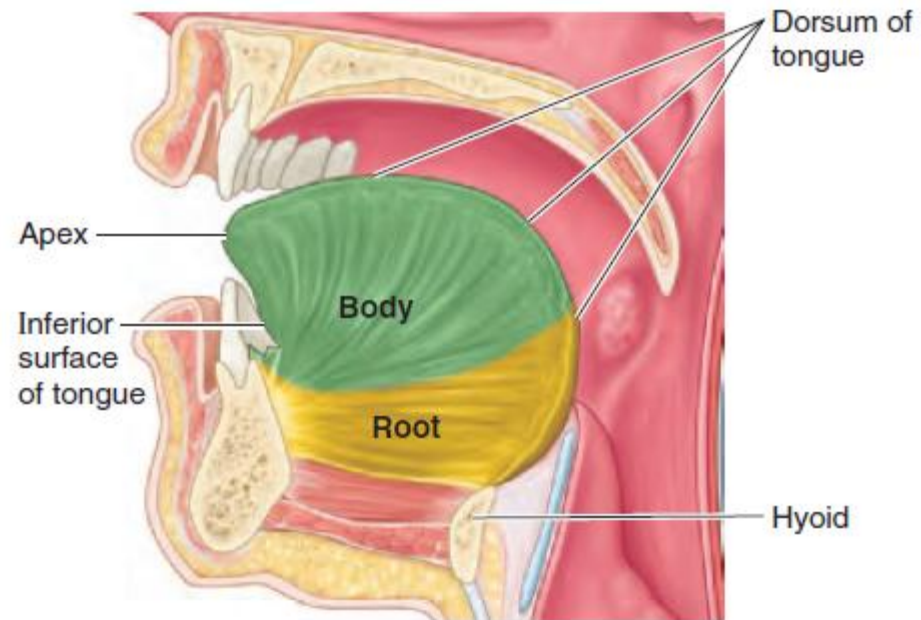


Tongue - Parts

- Mobile organ involved with
 - taste
 - mastication
 - deglutition
 - articulation

Main parts

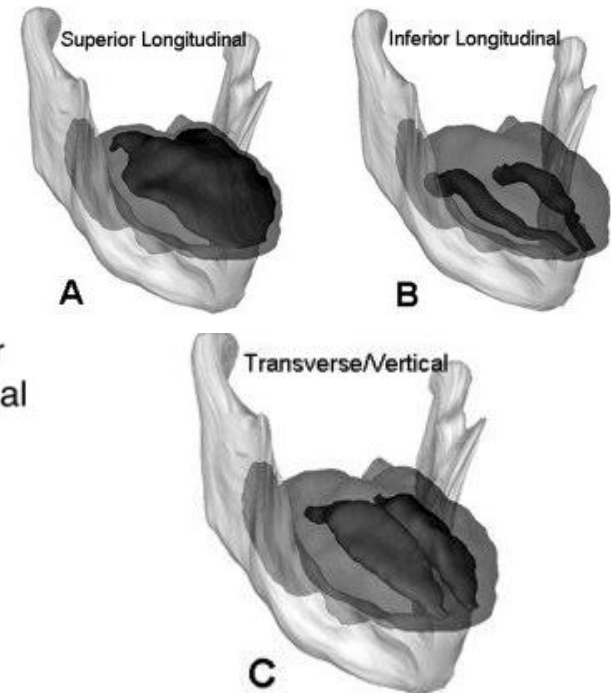
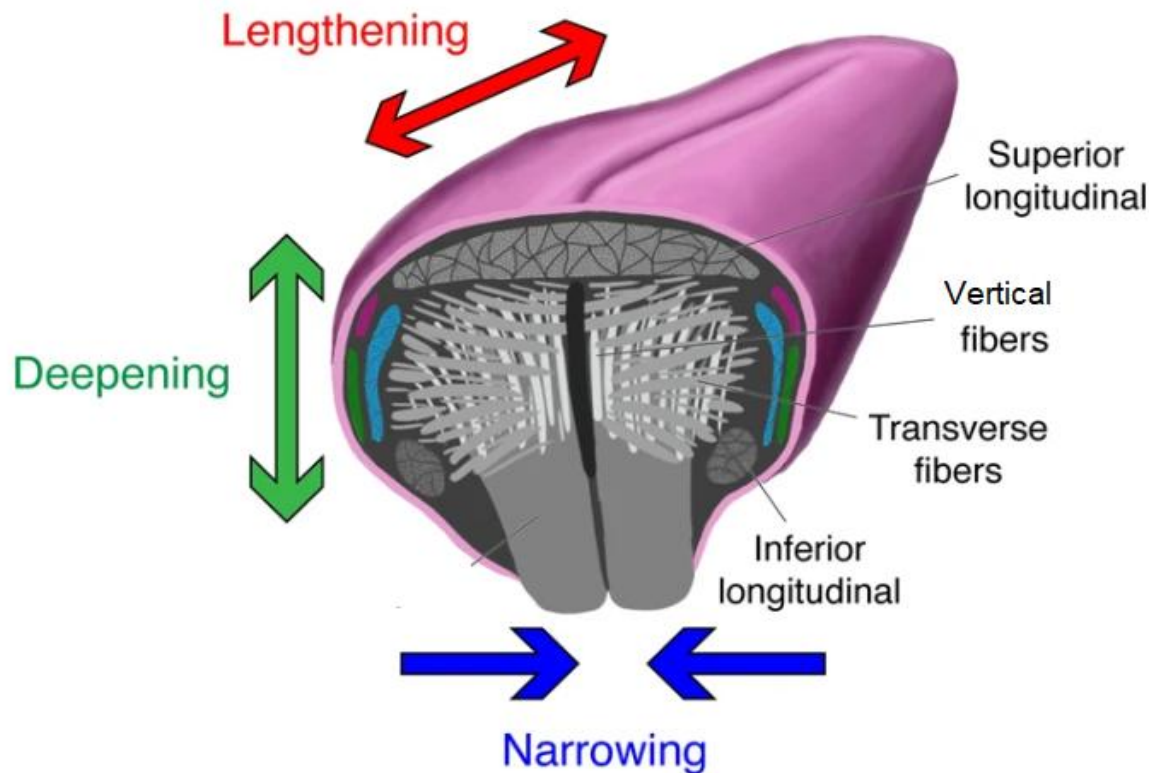
- Dorsum: dorsal surface
- Root: posterior third
 - rests on floor of mouth
- Body: anterior two-thirds
- Apex: tip of the tongue
 - rests against incisors



(A) Median section of mouth

Intrinsic Tongue Muscles

- Alter the tongue's shape
 - Longitudinal: curl and uncurl apex and edges
 - Transverse: lengthen or narrow
 - Vertical: flatten/deepen or round
- Innervated by XII



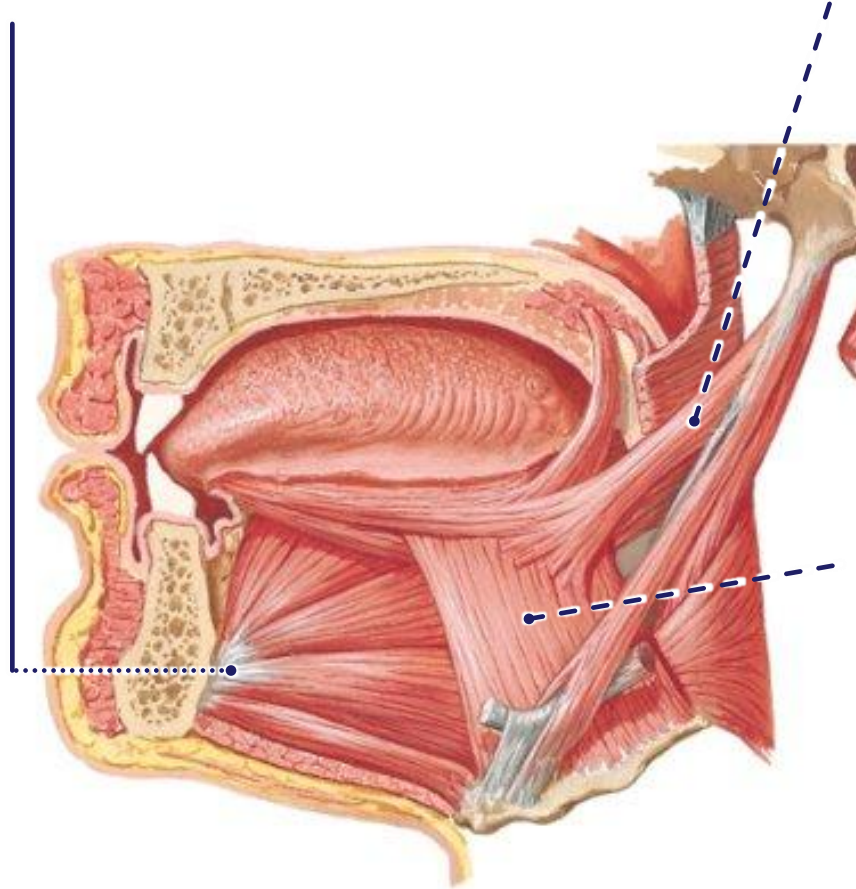
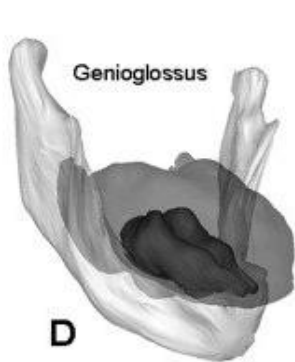
DOI: 10.1002/ar.22711

DOI: 10.1038/s41598-020-64935-z

Extrinsic Tongue Muscles

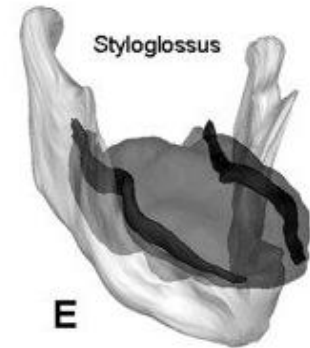
Genioglossus: innervated by XII

- depresses central part of tongue
- protrudes anterior tongue beyond oral fissure



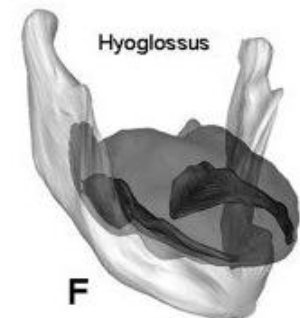
Styloglossus: innervated by XII

- retracts posterior tongue

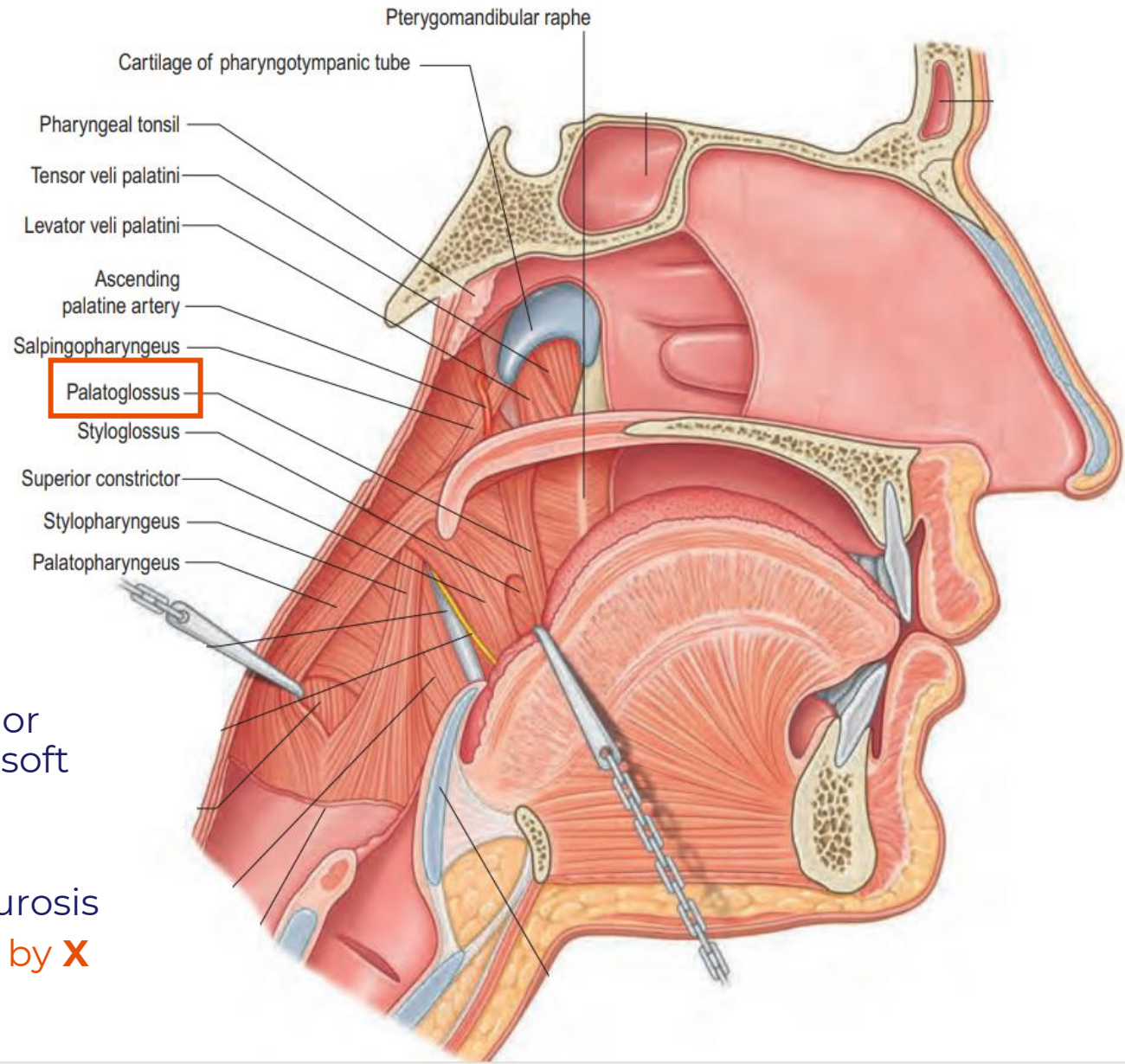


Hyoglossus: innervated by XII

- depresses tongue



Palatoglossus

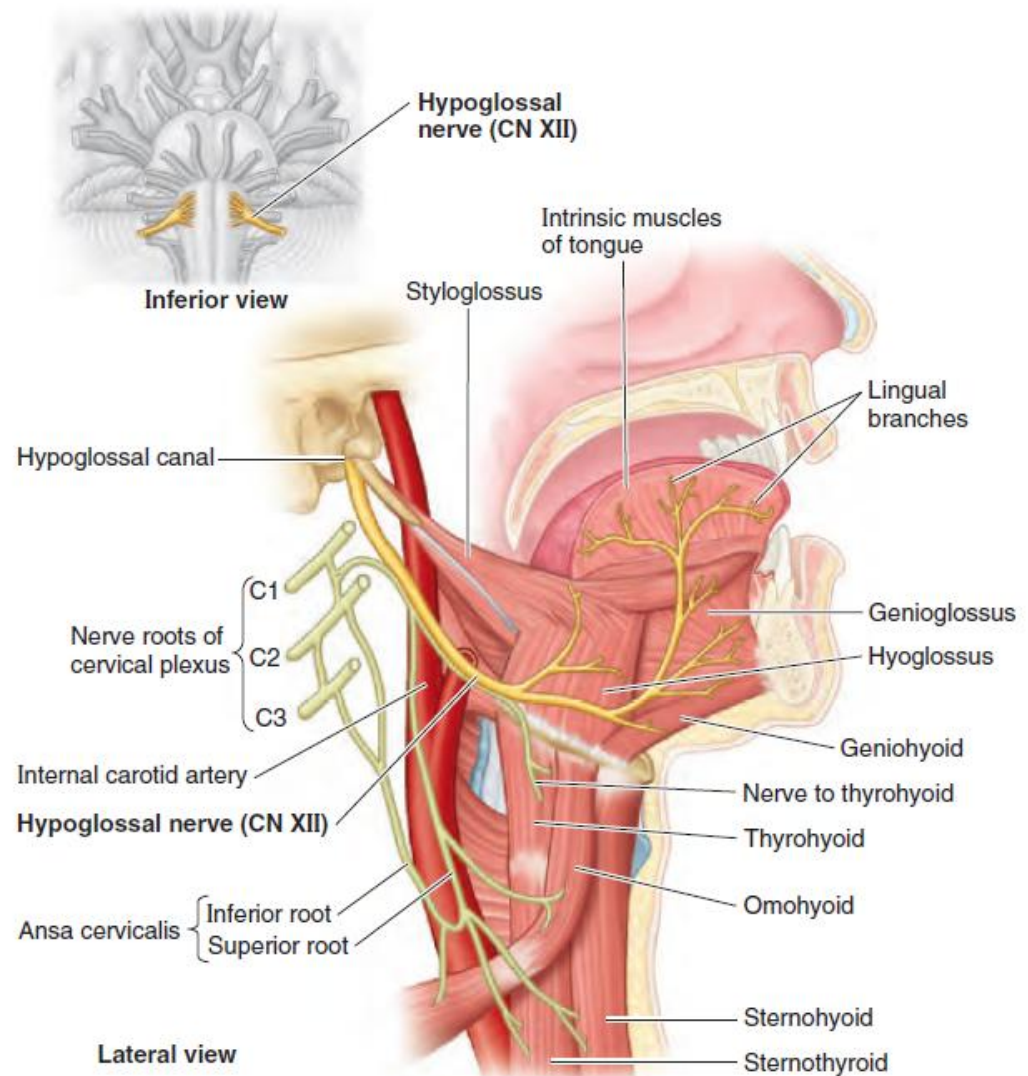


Palatoglossus

- elevates posterior tongue toward soft palate
- originates from palatine aponeurosis
 - innervated by X

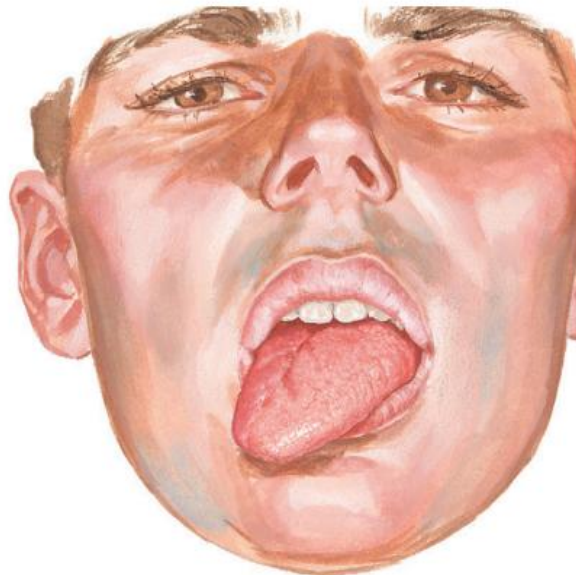
Hypoglossal nerve (XII) - course

- Fibers originate from hypoglossal nucleus
- Leaves skull through the hypoglossal canal of the *occipital bone*
- Passes deep to mandible to enter the tongue
- Extracranially is joined by **cervical plexus branches**
 - somatic motor from C1, C2
 - **Superior root of ansa cervicalis branches** supplies **infrahyoid muscles**
 - somatic sensory from C2
 - **Meningeal branch** returns to cranium through hypoglossal canal
 - innervates floor and posterior wall of **posterior cranial fossa**

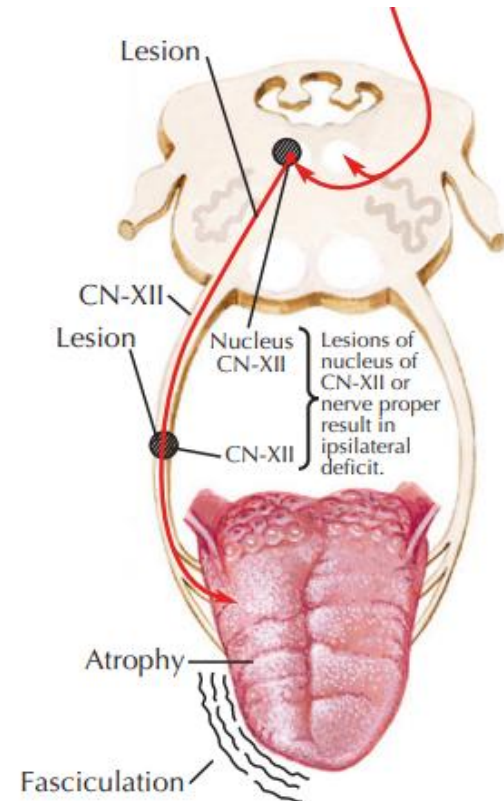


Hypoglossal nerve - injury

- Supplies all intrinsic & extrinsic tongue muscles **except** palatoglossus
- Acute XII lesion paralyzes **ipsilateral** half of tongue
 - elicited tongue protrusion shows apex deviating towards paralyzed side
 - due to unopposed action of genioglossus
- Paralyzed genioglossus tends to fall posteriorly and obstruct airway
 - Airway precautions are taken before general anesthesia
- Chronic injury can lead to tongue **atrophy**

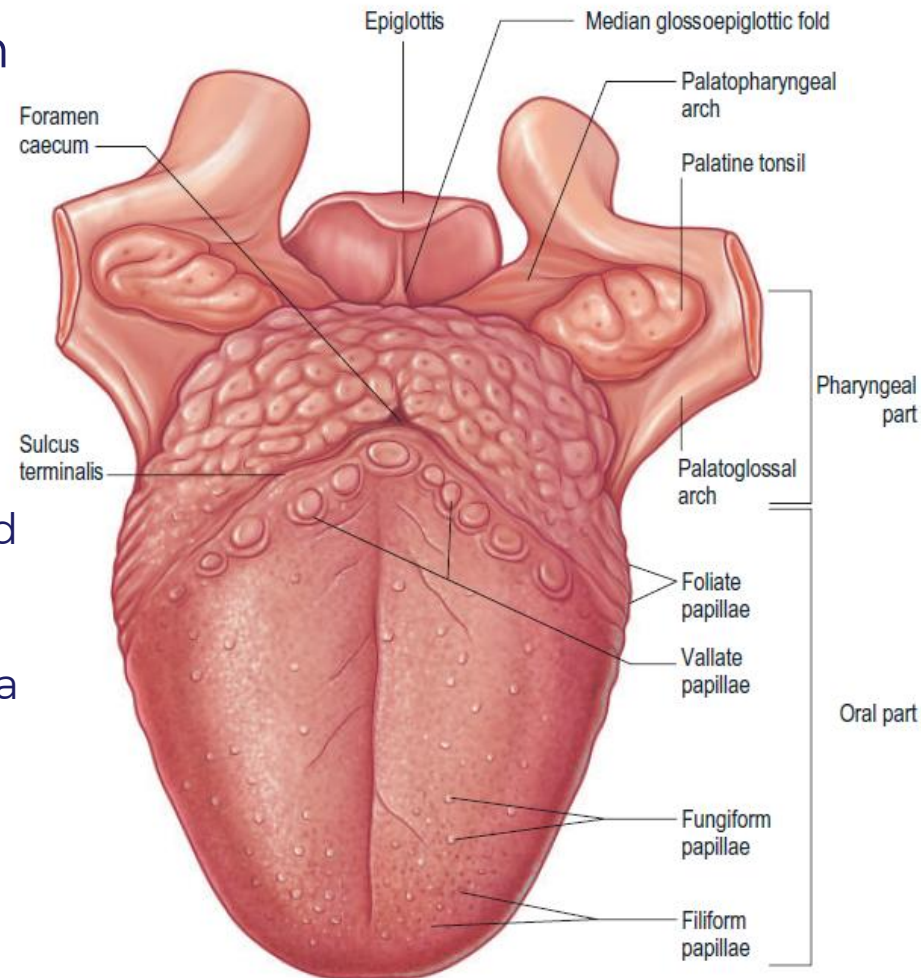


Right hypoglossal nerve paralysis:
tongue deviates *toward* paralyzed
side when protruded



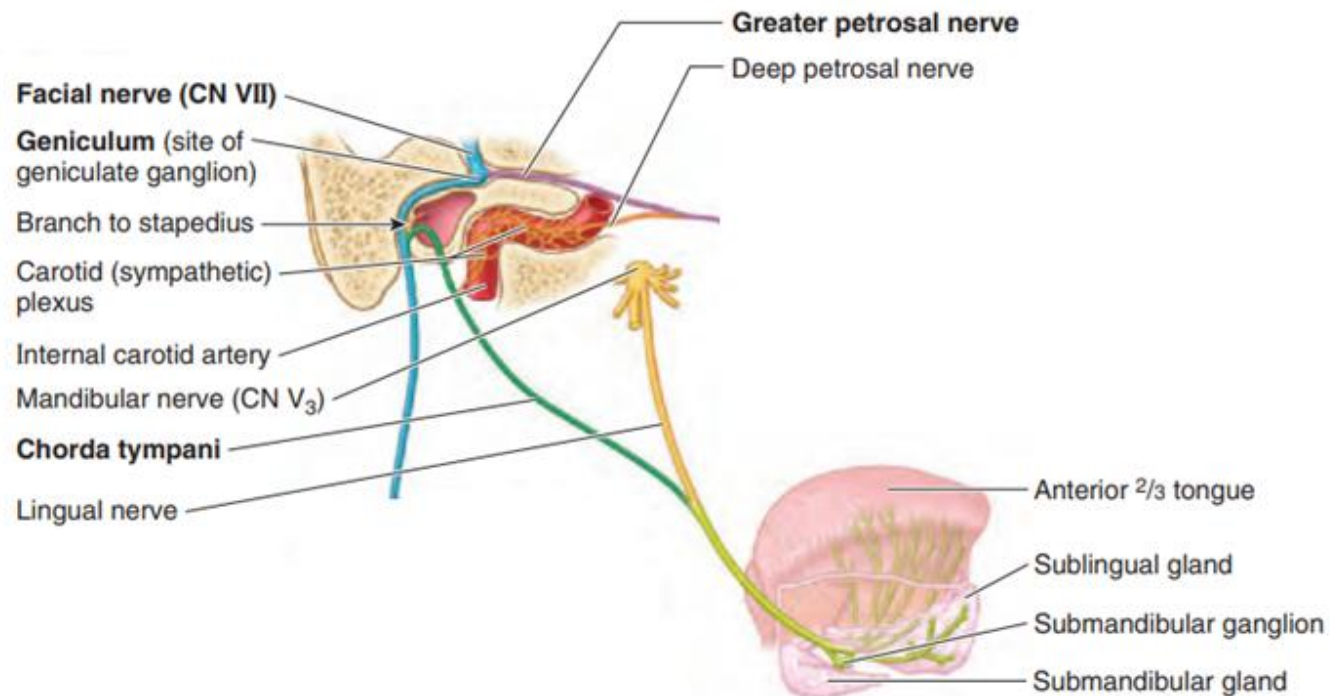
Tongue – Surface Anatomy

- *Terminal sulcus*
 - forms margin between oral and pharyngeal cavities
- Numerous lingual papillae on anterior tongue
 - Vallate:
 - directly anterior to terminal sulcus
 - Filiform: **sensitive to touch**
 - begin parallel to the terminal sulcus
 - Fungiform
 - most numerous at apex and margins
 - Foliate
 - small lateral folds of mucosa
- All papillae **except filiform** contain taste receptors in taste buds



Tongue - Sensory Overview

- Functions in feeding, tasting, swallowing, airway separation
- Carries general (touch, pain, temperature) and special (taste) sensation via
 - **Lingual nerve (V₃)** for general sensory
 - **Chorda tympani (VII)** for taste (anterior two-thirds of tongue)
 - **Lingual branch of IX** for general sensation & taste (posterior tongue)
 - **Internal laryngeal nerve (X)** for general sensation & taste (posterior tongue)



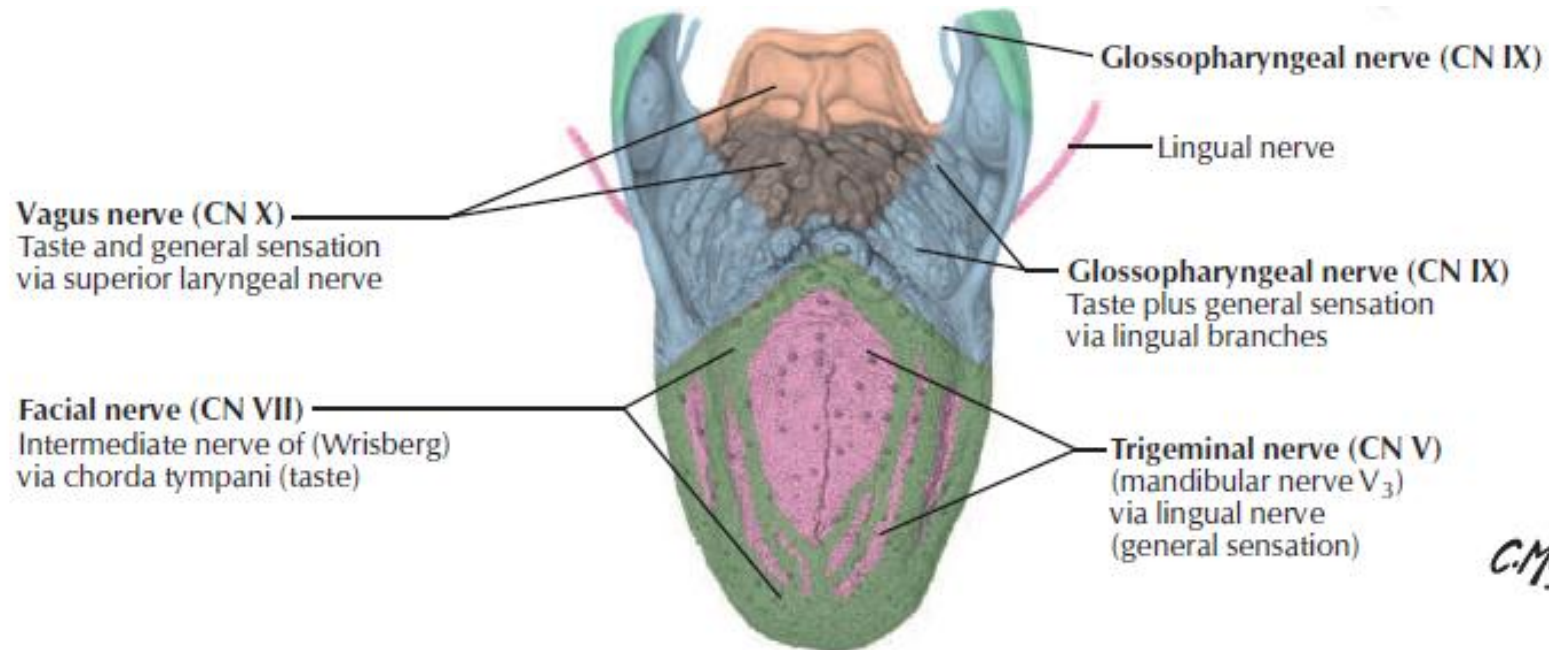
Tongue - Sensory Innervation

Carries General (touch, pain, temperature) and Special (taste) sensation

- General sensory: lingual nerve (V_3)
- Taste on anterior two-thirds: chorda tympani (VII)

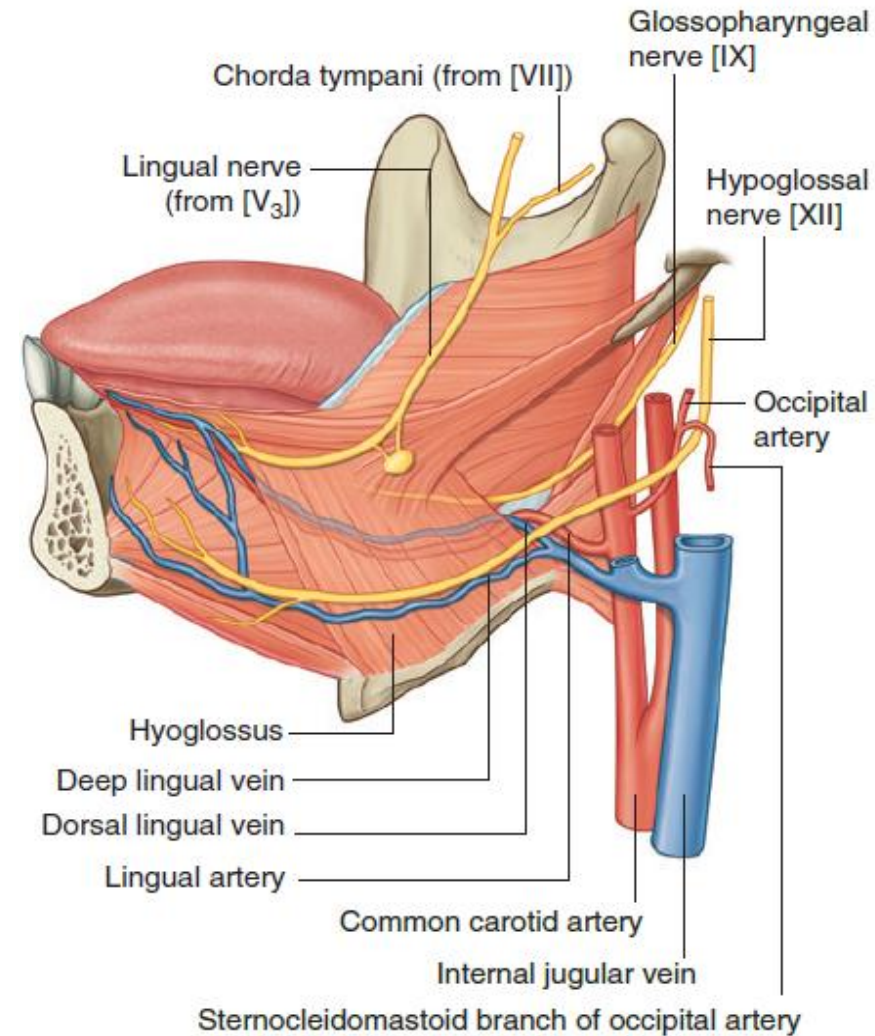
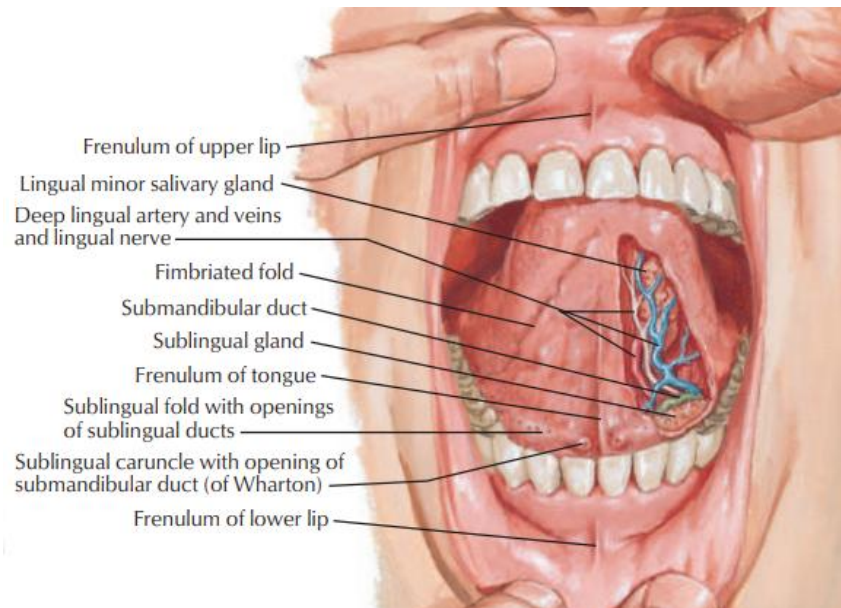
Both:

- internal laryngeal nerve (X)
- lingual branch of IX



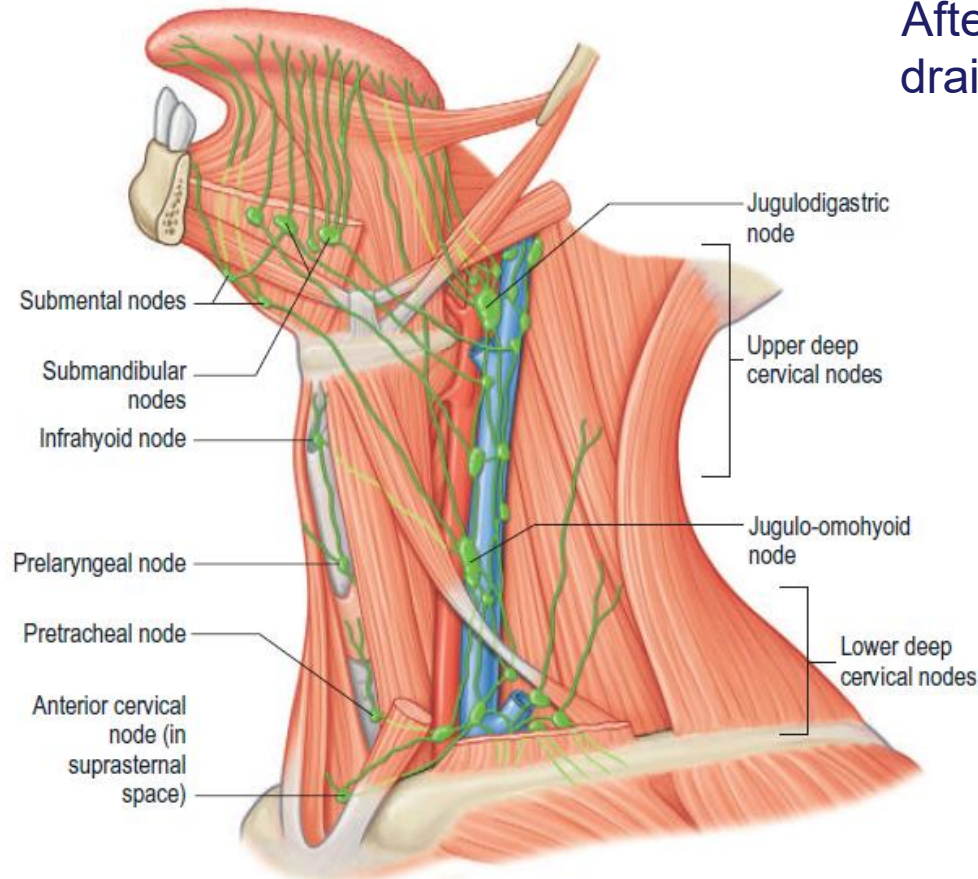
Tongue – Blood Supply

- *Lingual artery*
 - from the external carotid
- **Dorsal lingual vein**
 - follows lingual artery
- **Deep lingual veins**
 - travel with XII on external surface of tongue
 - visible under tongue

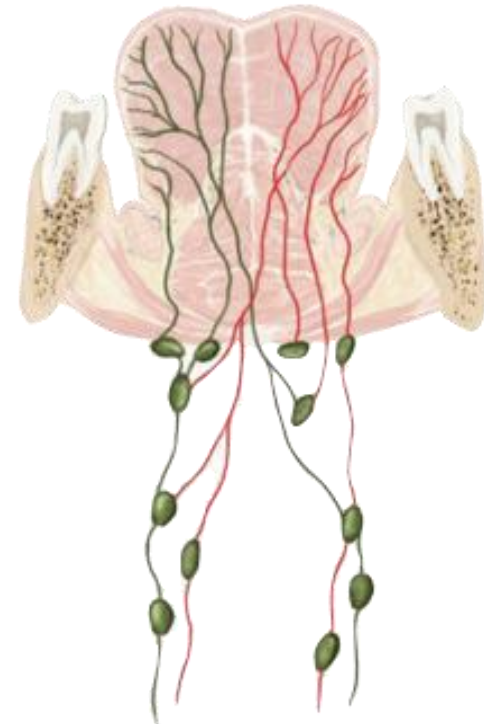


Tongue – Lymphatic Drainage

- Tongue tip: drains to **submental nodes**
- Lateral tongue: drains to **inferior deep cervical nodes**
- Remainder drains to **superior deep cervical nodes**

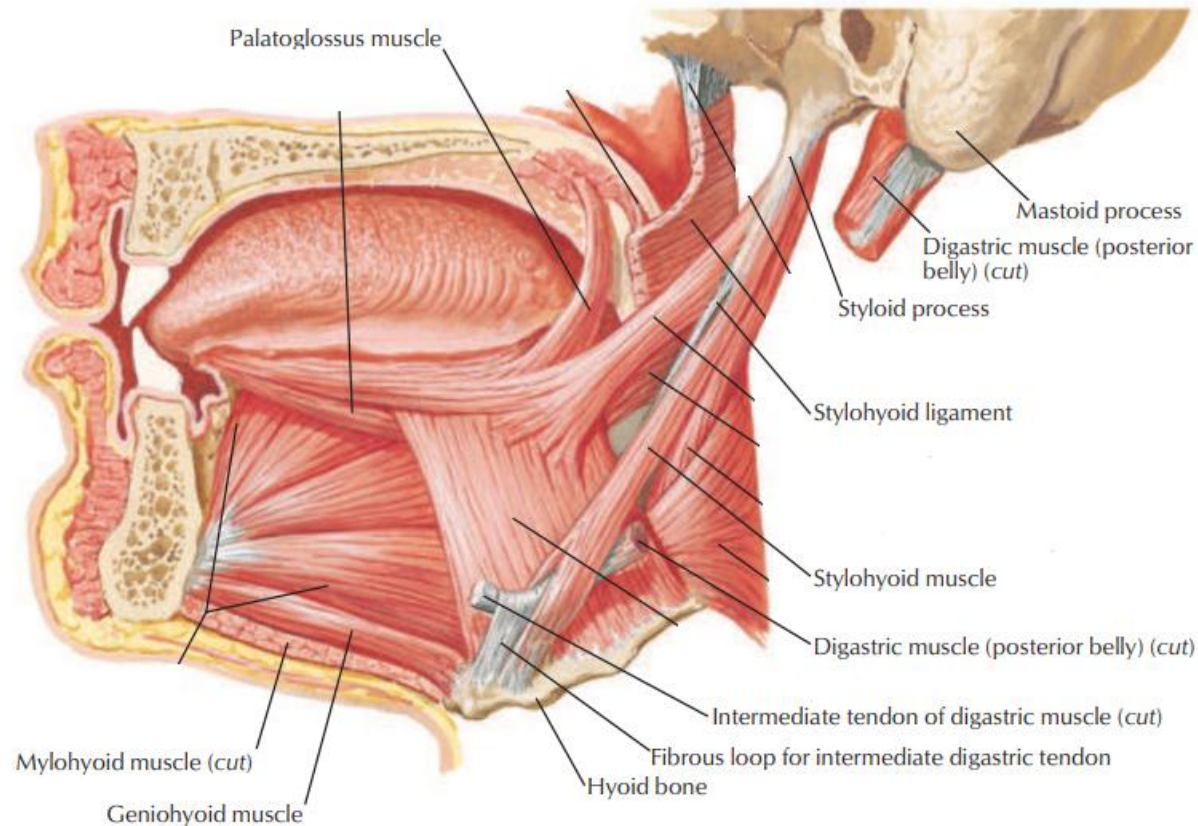


After descending under the mucosa, drainage moves to distributed nodes and can pass bilaterally



Oral Floor & Suprahyoid Muscles

- *Hyoid bone* is elevated and moved forward during swallowing
- Suprahyoid muscles
 - Oral floor: **Mylohyoid** and **Geniohyoid**
 - Digastric (**anterior** and **posterior** belly) and **Stylohyoid**

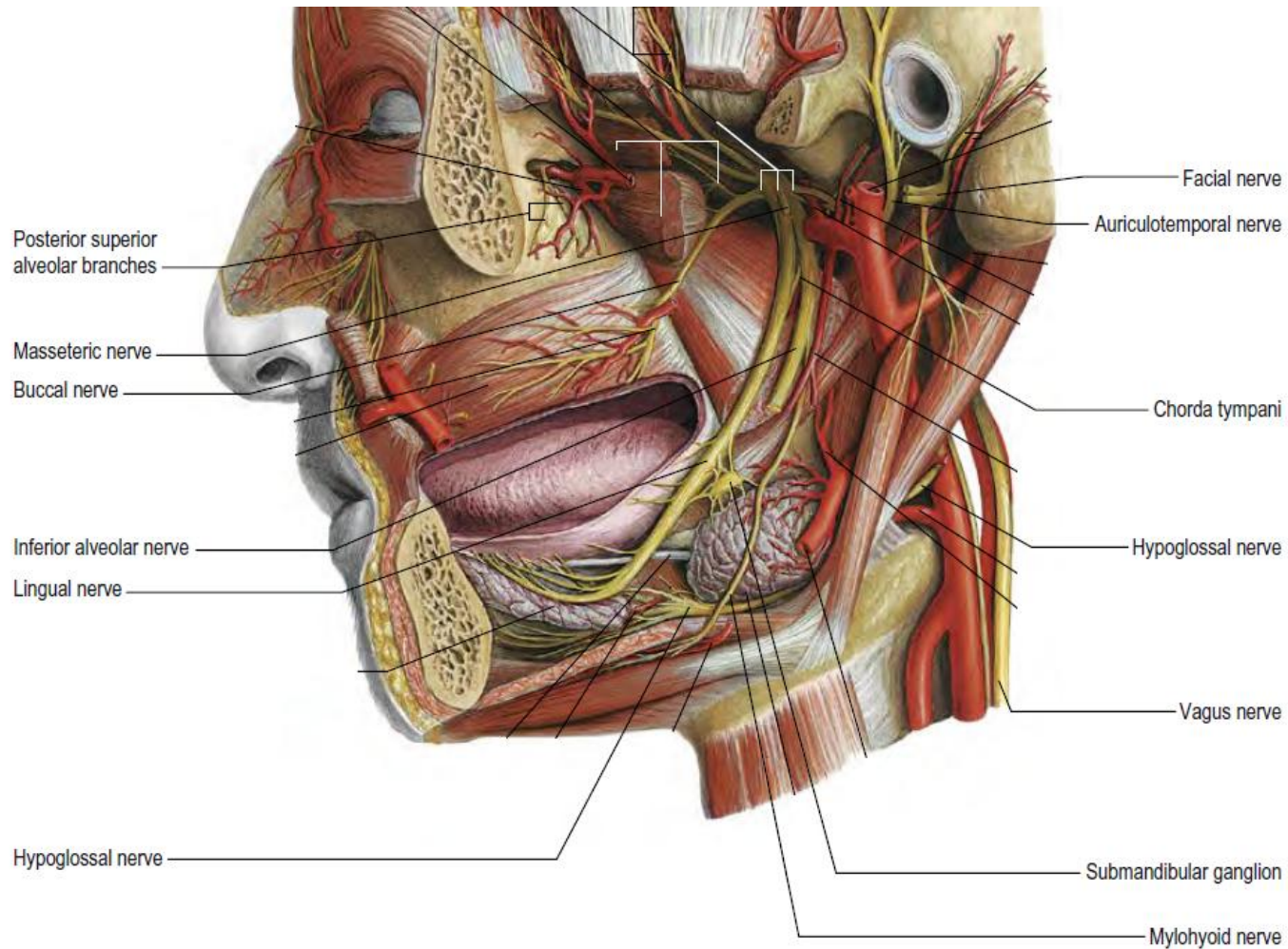


Lateral view (left side of mandible removed)

Innervation of Suprahyoid Muscles

Mylohyoid nerve supplies:

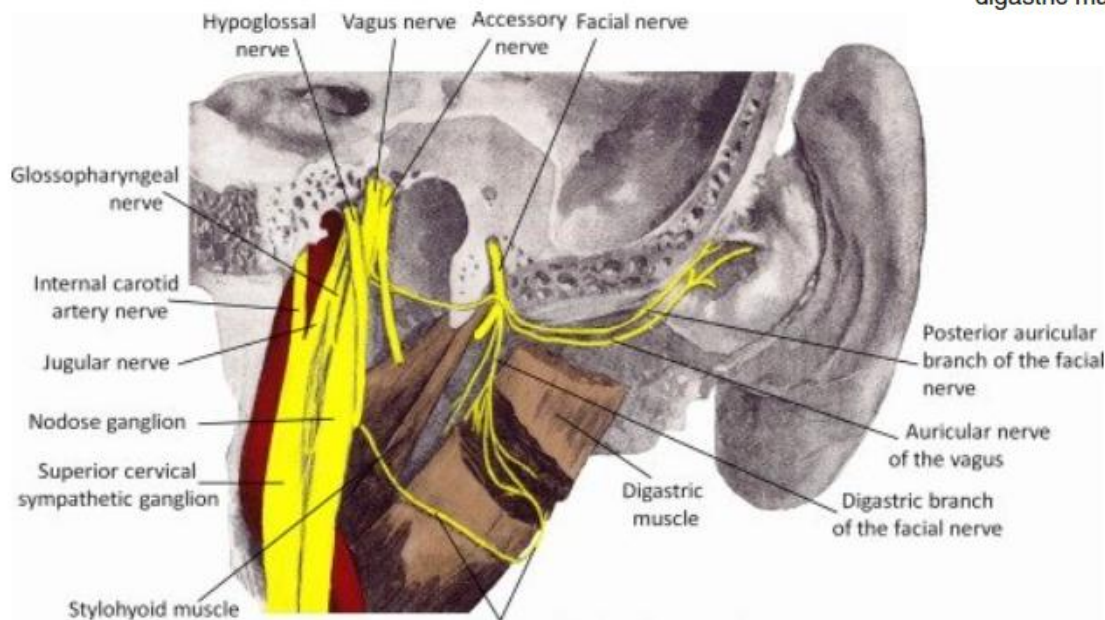
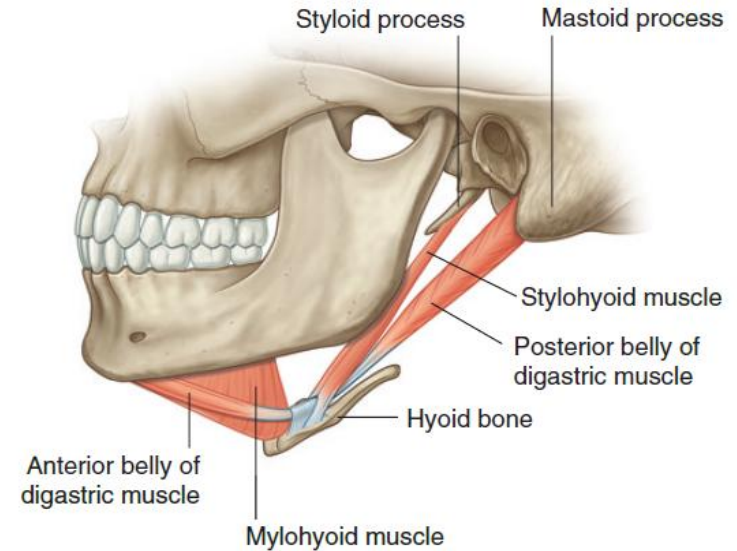
- Mylohyoid
- Anterior belly of digastric



Innervation of Suprahyoid muscles

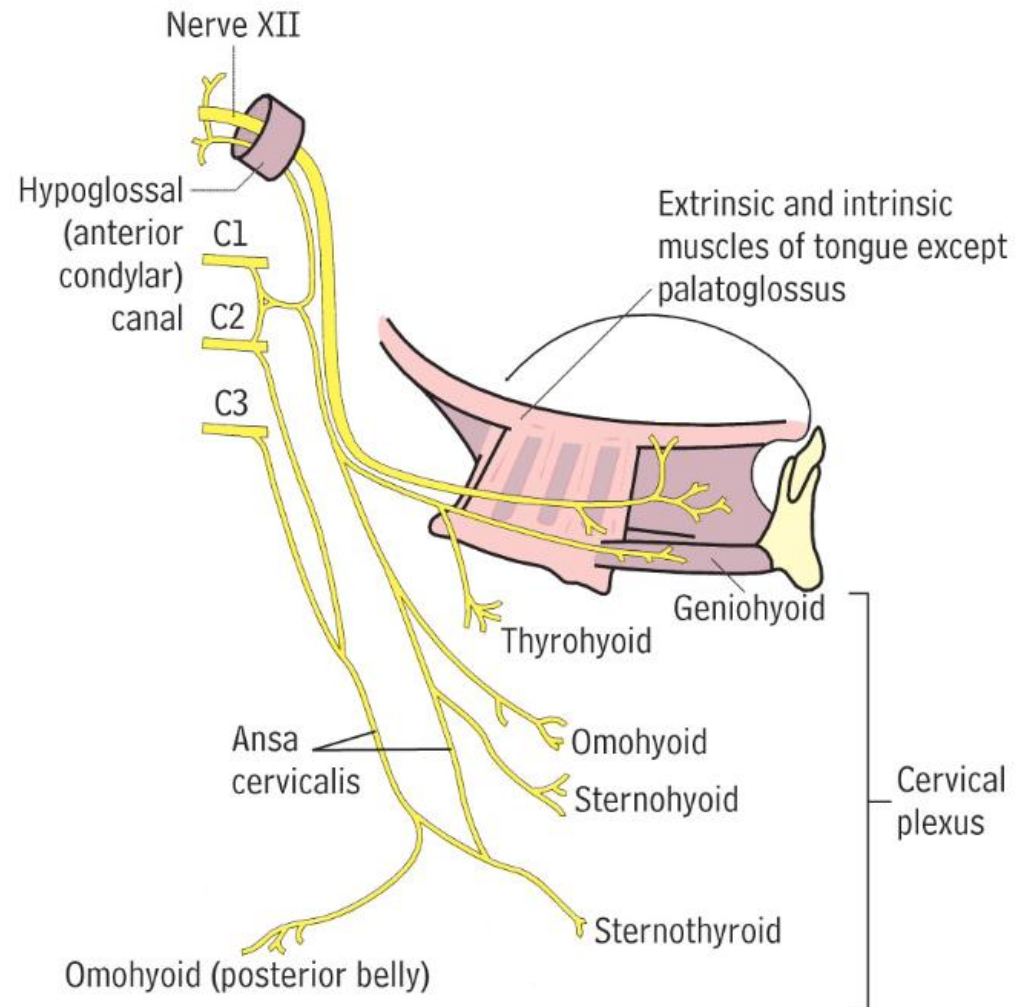
Facial nerve supplies

- Posterior digastric
- Stylohyoid



Innervation of Suprahyoid Muscles

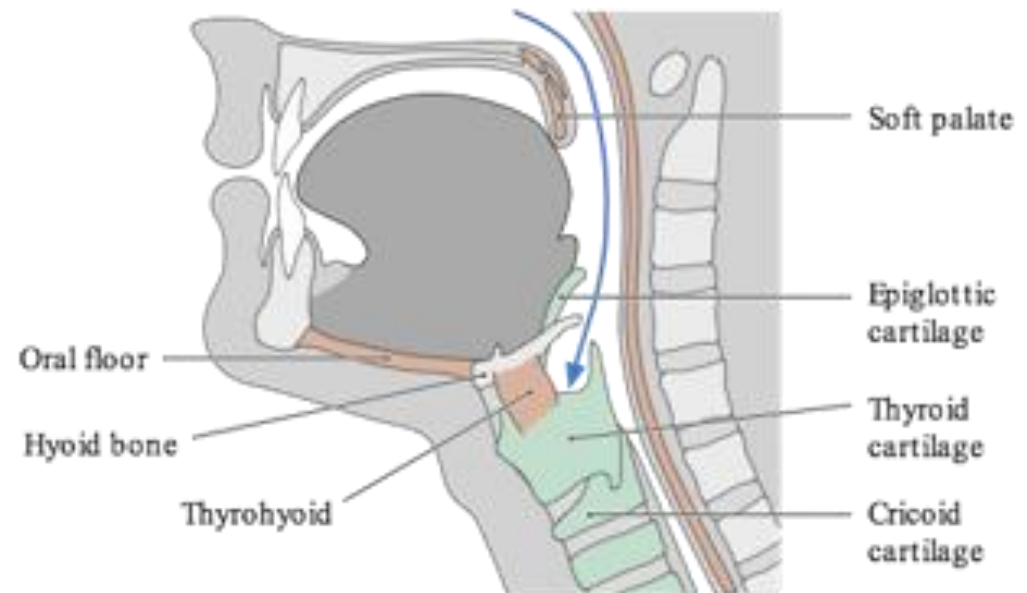
- Geniohyoid
 - supplied by a branch from the **ventral ramus of spinal nerve C1**
 - travels with **XII**



Anatomy of Swallowing

Three phases to deglutition

1. *Voluntary initiation & transit (oral phase)*
2. *Reflex closure of the airway (pharyngeal phase)*
3. *Reflex transport (esophageal phase)*



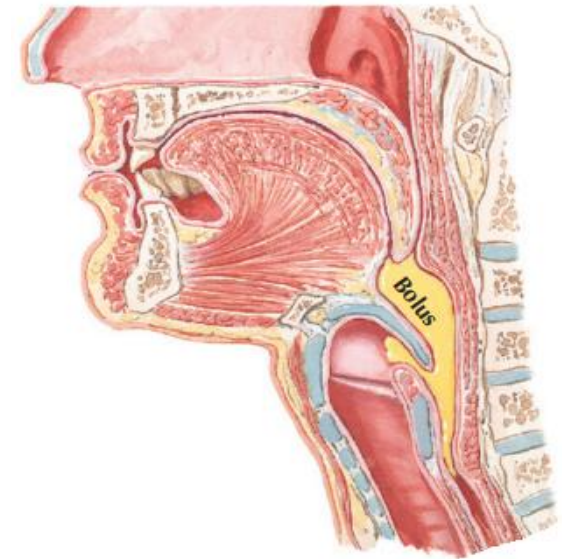
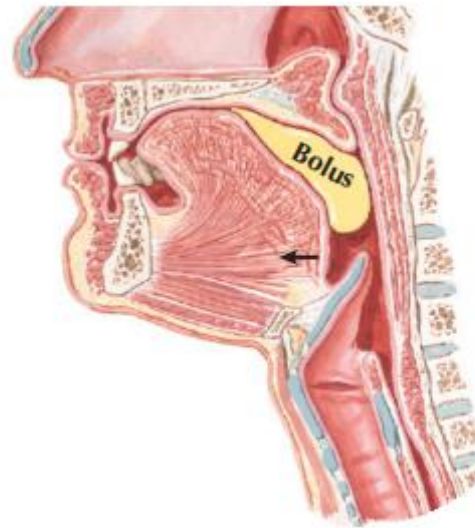
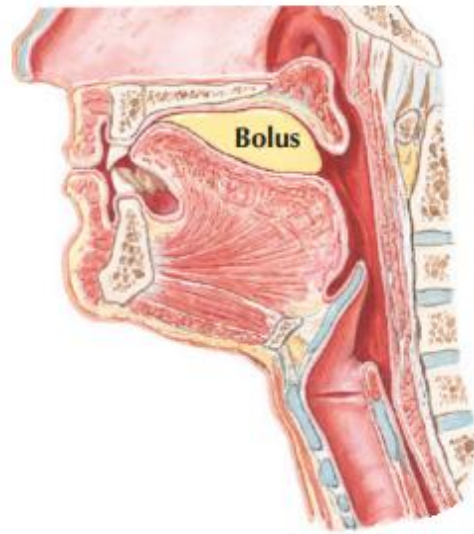
Initiation of swallowing

Anterior tongue raises

- genioglossus
- intrinsic tongue muscles

Soft palate lowers

- palatoglossus
- pharyngeal longitudinal muscles

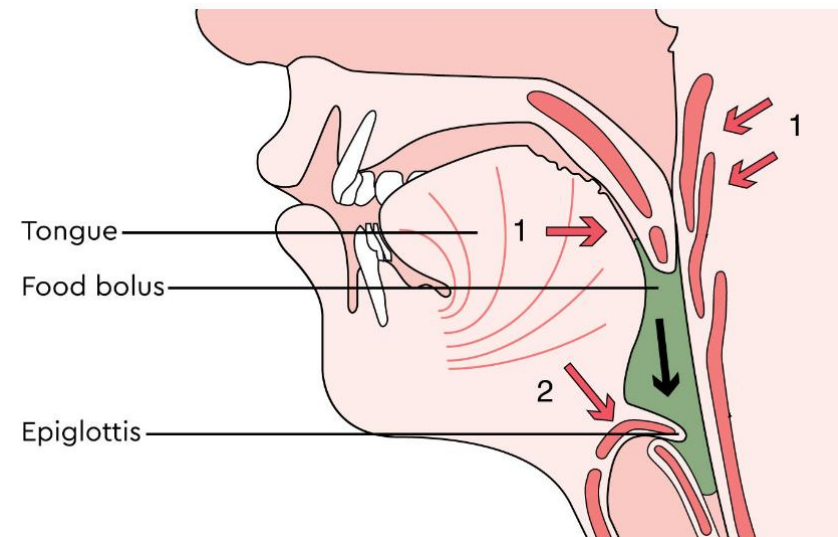
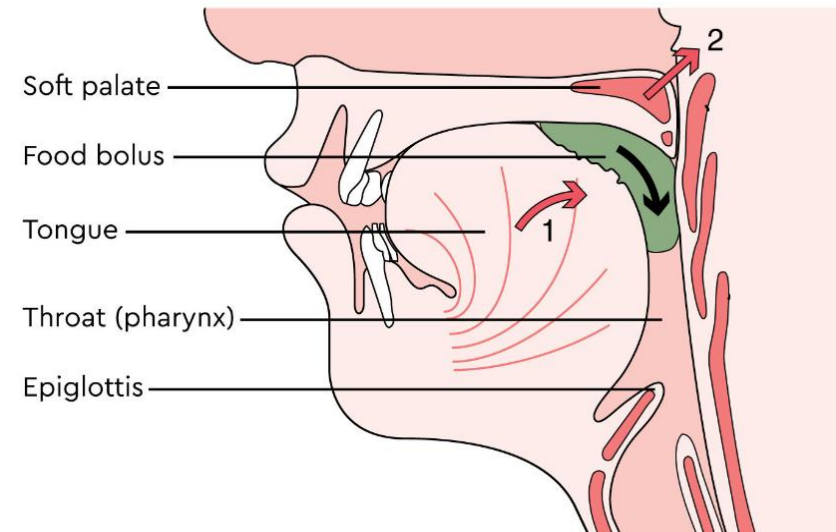


Posterior tongue retracts

- styloglossus

Oral Transit

- Soft palate muscles act as a valve
 - If **elevated** separates nasopharynx from oropharynx
 - If **depressed** closes off the oropharyngeal isthmus
- Innervated by vagus (X)
 - **except** tensor veli palatini (V3)

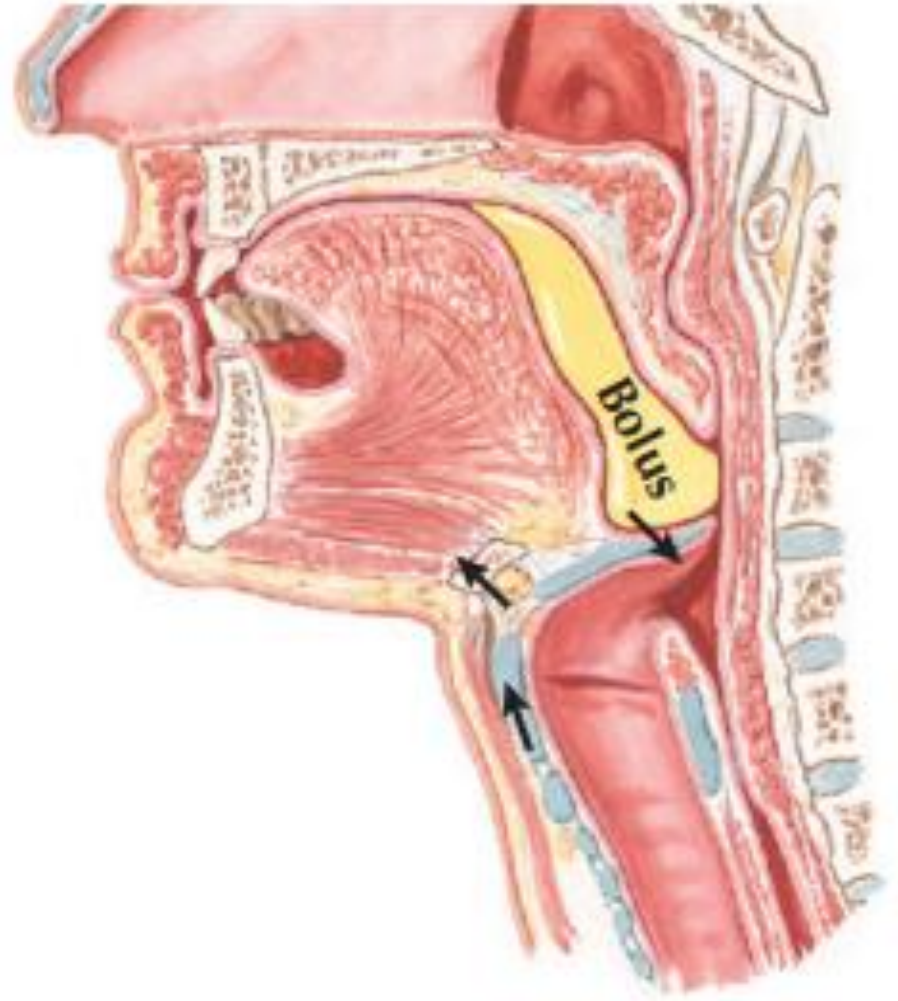


<https://www.macmillan.org.uk/cancer-information-and-support/head-and-neck-cancer/what-happens-when-you-chew-and-swallow>

Pharyngeal Phase

Reflex closure of the airway

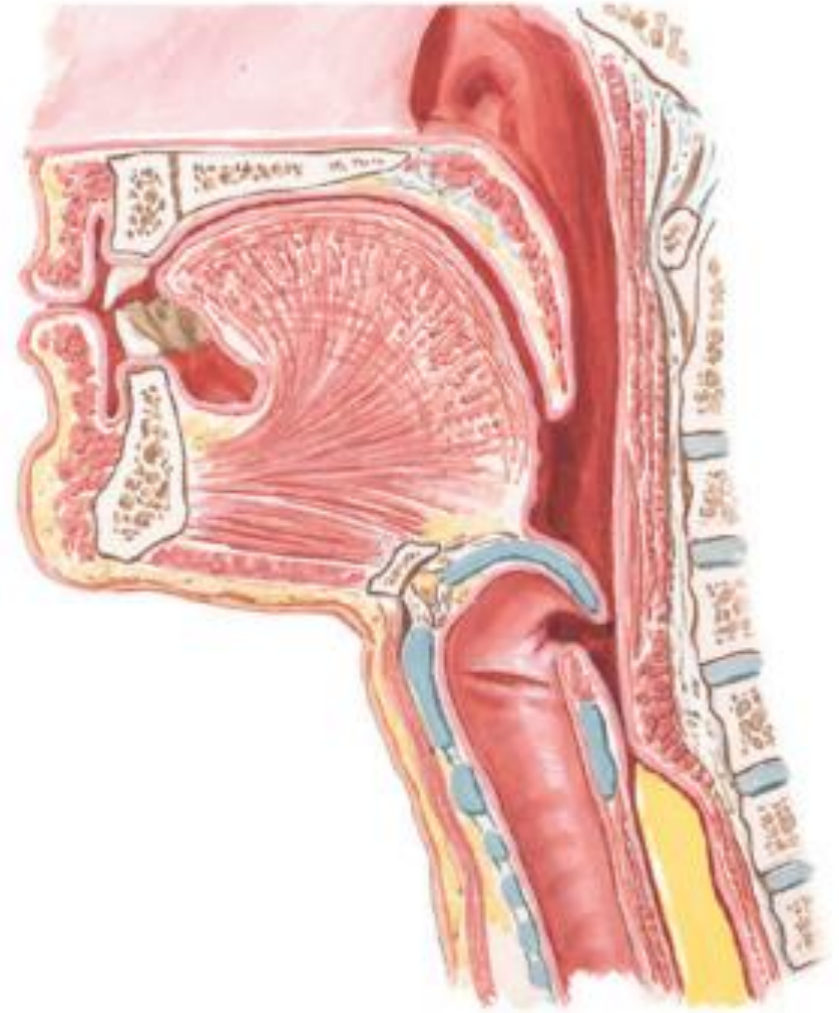
- Larynx is elevated
 - **mylohyoid**
 - **thyrohyoid**
 - **digastric**
- Laryngeal inlet is sealed
 - covered by **epiglottis**



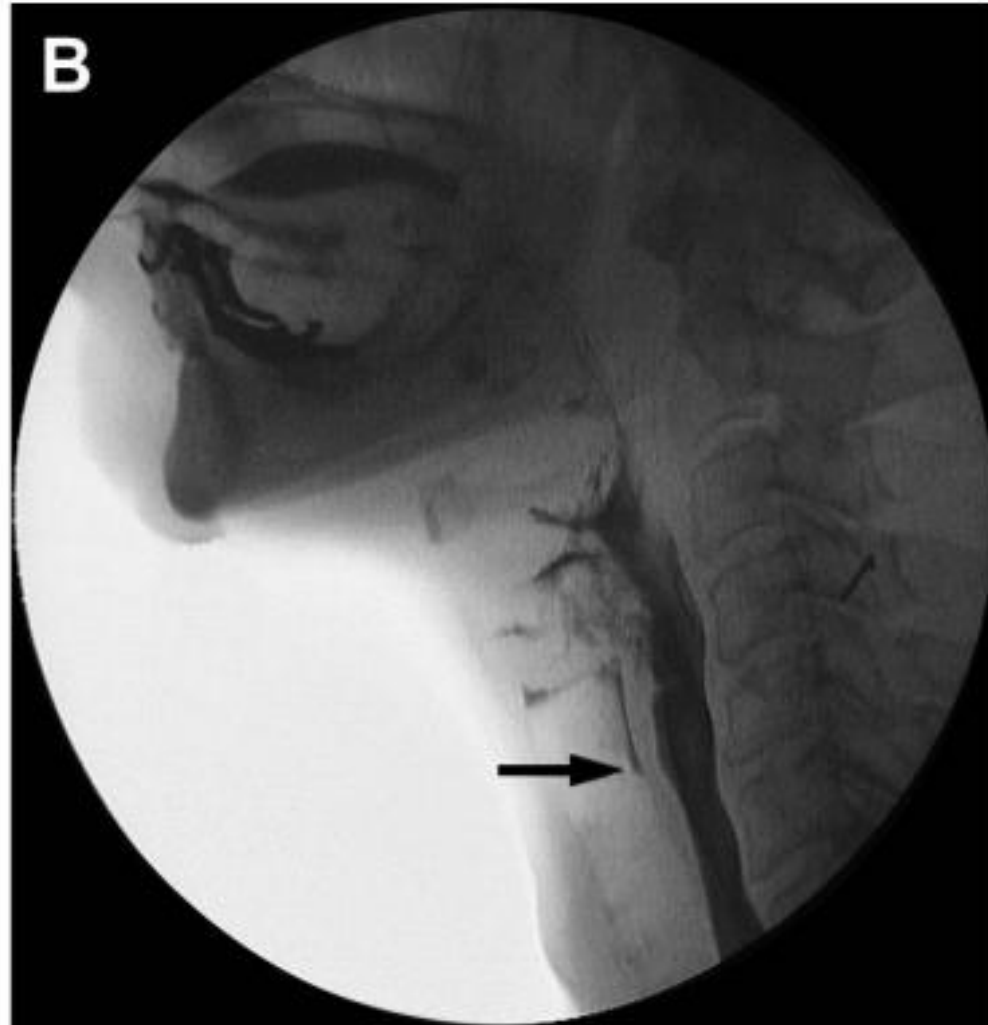
Esophageal Phase

Reflex transport

- Upper esophageal sphincter relaxes
- Sequential waves of contraction move the bolus down to the lower esophageal sphincter and stomach
- much more variable than other phases



Dysphagia & Videofluoroscopy



DOI: 10.1016/j.pmr.2008.06.001.

fin